

TRANSUNION INSIGHTS & AUTO- REVOLVING LOANS

DATA SCIENCE INTERN PROJECT

Kylie Alberts
2025



American
AgCredit



ABOUT ME

1

Grew up in San Francisco Bay Area

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2

B.S. in Neuroscience at Union College



ABOUT ME

1

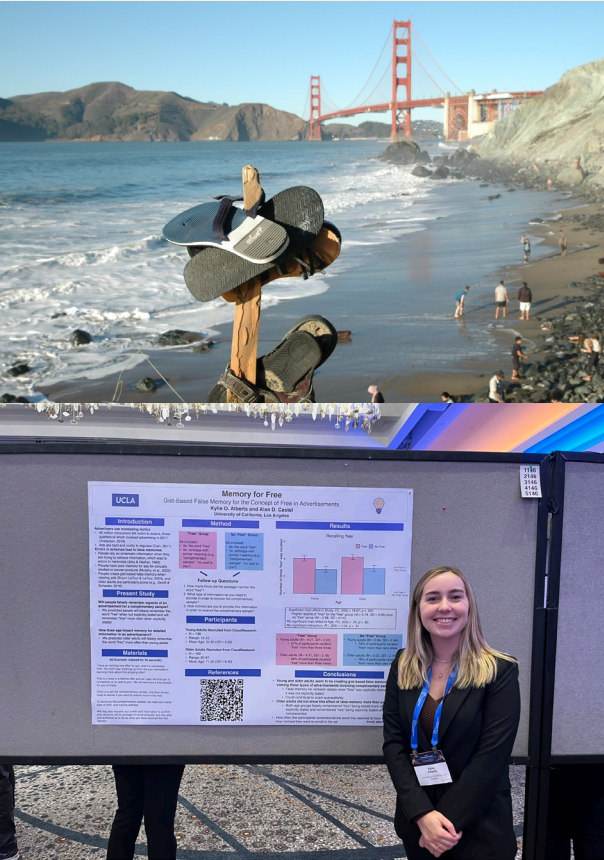
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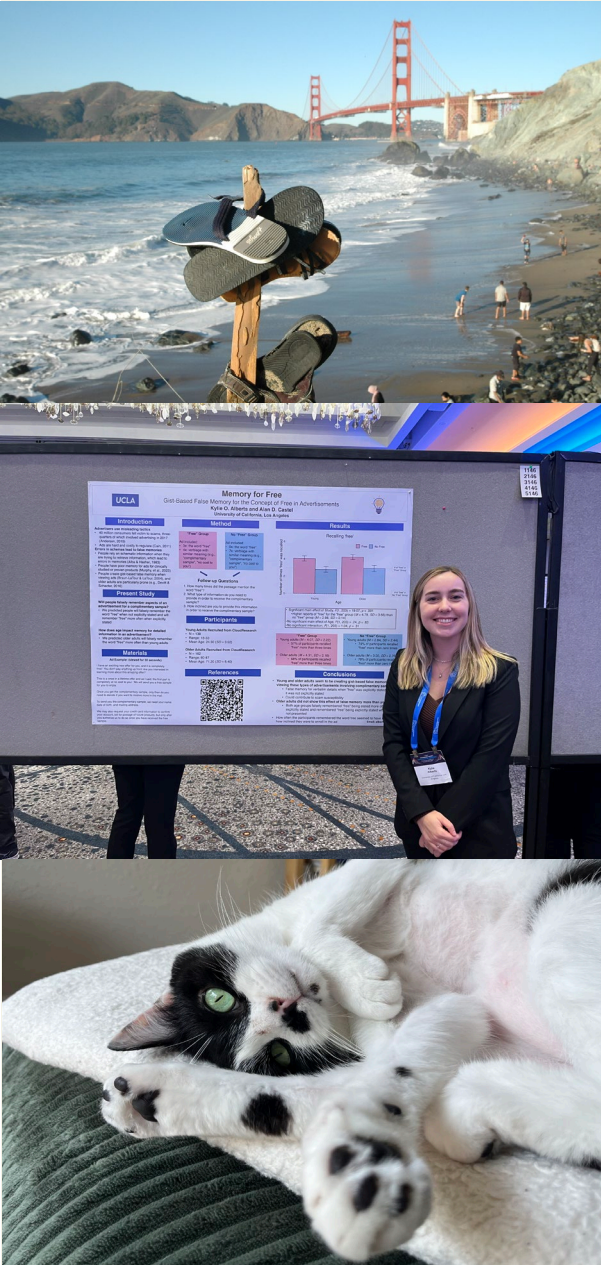
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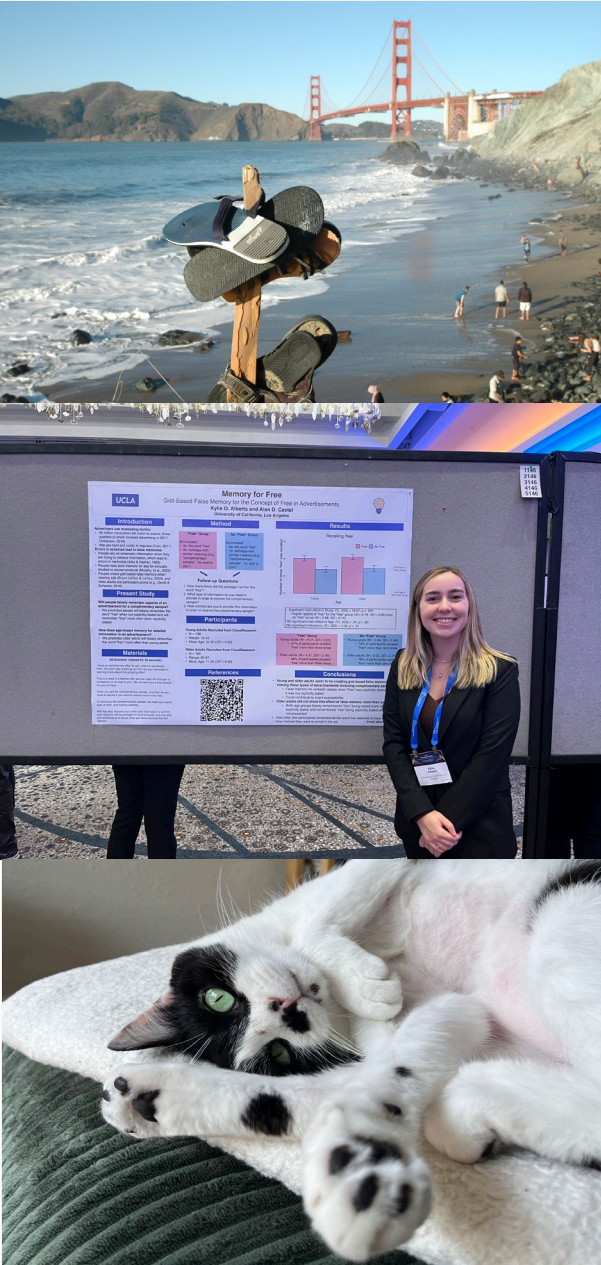
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Currently living in Los Angeles

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Pursuing a career in Data Science!



INTERNSHIP DAY-TO-DAY



Los Angeles

Manager: Minet Polsinelli

Initial Research

- First to explore TransUnion Data
- Gathered initial insights into goals and expectations of the project
- Explored potential tools to implement

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- Used tools such as RStudio and Data Robot to remove unnecessary variables and enhance formatting of the data
- Combined internal data with TransUnion data

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Data Analyses

- Analyzed data for variable importance
- Created visualizations of the data
- Used these analyses to inform modeling decisions



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What I learned: 1) techniques for cleaning and analyzing credit data 2) new packages and codes in RStudio, and tools such as Data Robot 3) applications of data science in farm credit 4) And more!

SUMMER PROJECT



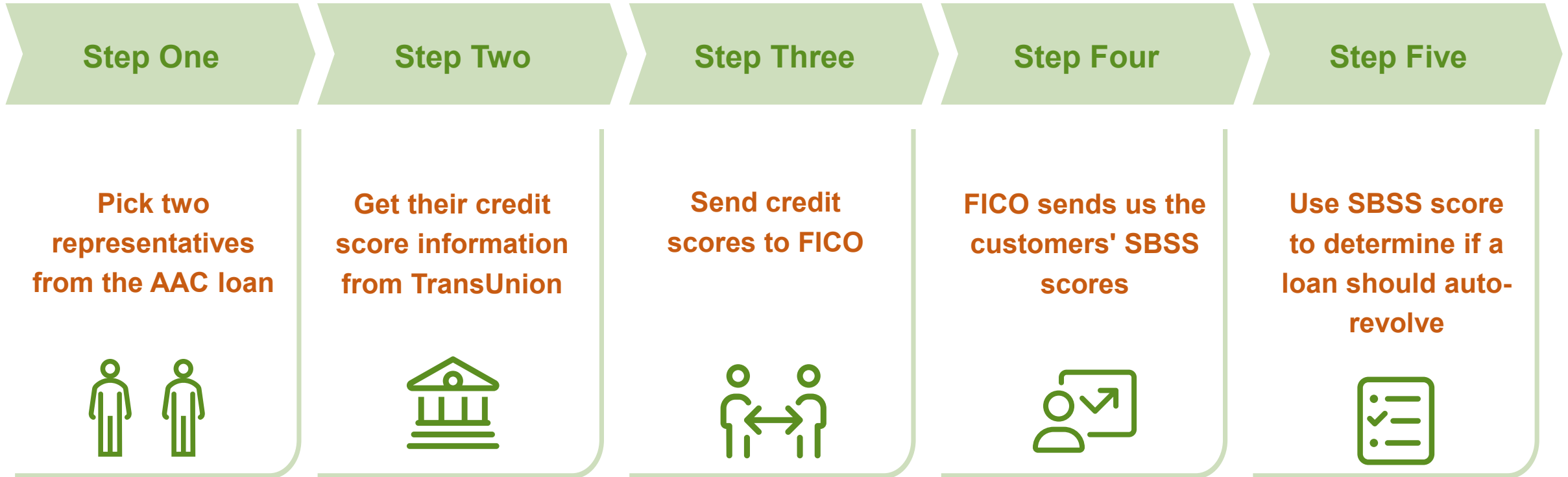
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Main Goal: To explore TransUnion data in combination with our internal data to see how we can leverage the TransUnion data to determine whether to auto-revolve a loan or not



CURRENT PROCESS



METHODS

1. Determined which variables predict a Bad Loan

- Ran analyses on revolving loans to see what variables are important in predicting Bad Loans
- Spoke with stakeholders to determine qualitative measures

2. Investigated variables with high variable importance further

- Created graphs and tables looking at thresholds for what could predict bad loans

3. Recommended a model

- Chose model that could be created to inform whether a loan should revolve or not



DATA SPECIFICS

Time Period

- 9/01/2019 thru 5/31/2025

Data Types

- **Internal Data**
 - FlatShareBalance
 - Customer loan level data (revolving loans only)
 - **TransUnion Data**
 - Used for credit reporting and has information on borrowers
 - SoftPullCustomerInfo
 - SoftpullCompleteCustomerActivity
- Ex: FICO Score, SBSS Score, Credit Usage

Target Variable

- **Bad Definition**
 - Any Litigation Code
 - (Bankruptcy, etc.)
 - Ever Non-Accrual (includes PD 12+)
 - Ever Charge-Off
 - Days Past Due > = 60
 - 60 days past due (ever/current)

VARIABLE IMPORTANCE- TRANSUNION VARIABLES

Average
account age

Number
of Accounts

SBSS Score

FICO Score

Utilization
Ratio

Transaction Year

High
Utilization
Flag

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account age

Number
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SBSS Score

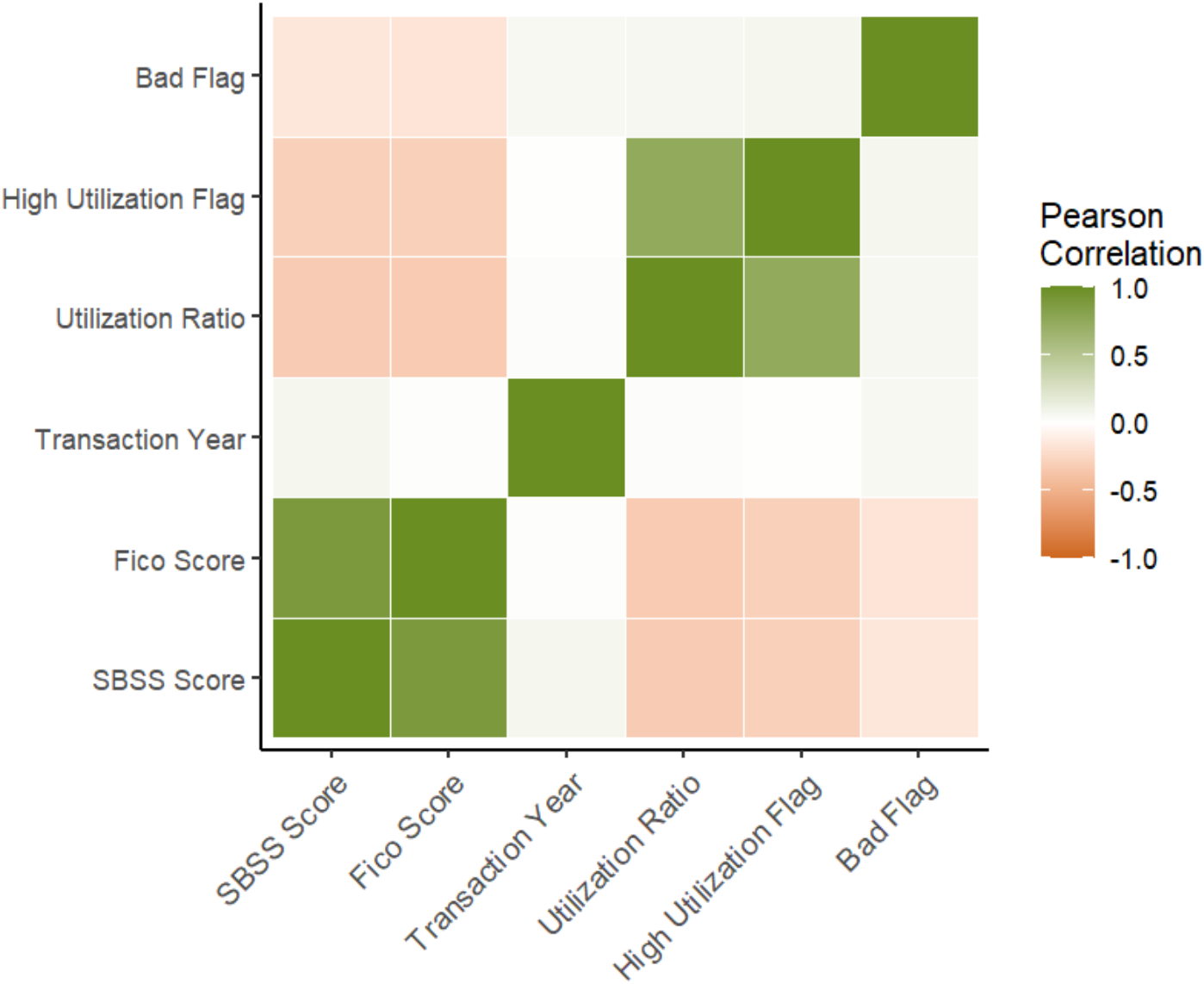
FICO Score

**Utilization
Ratio**

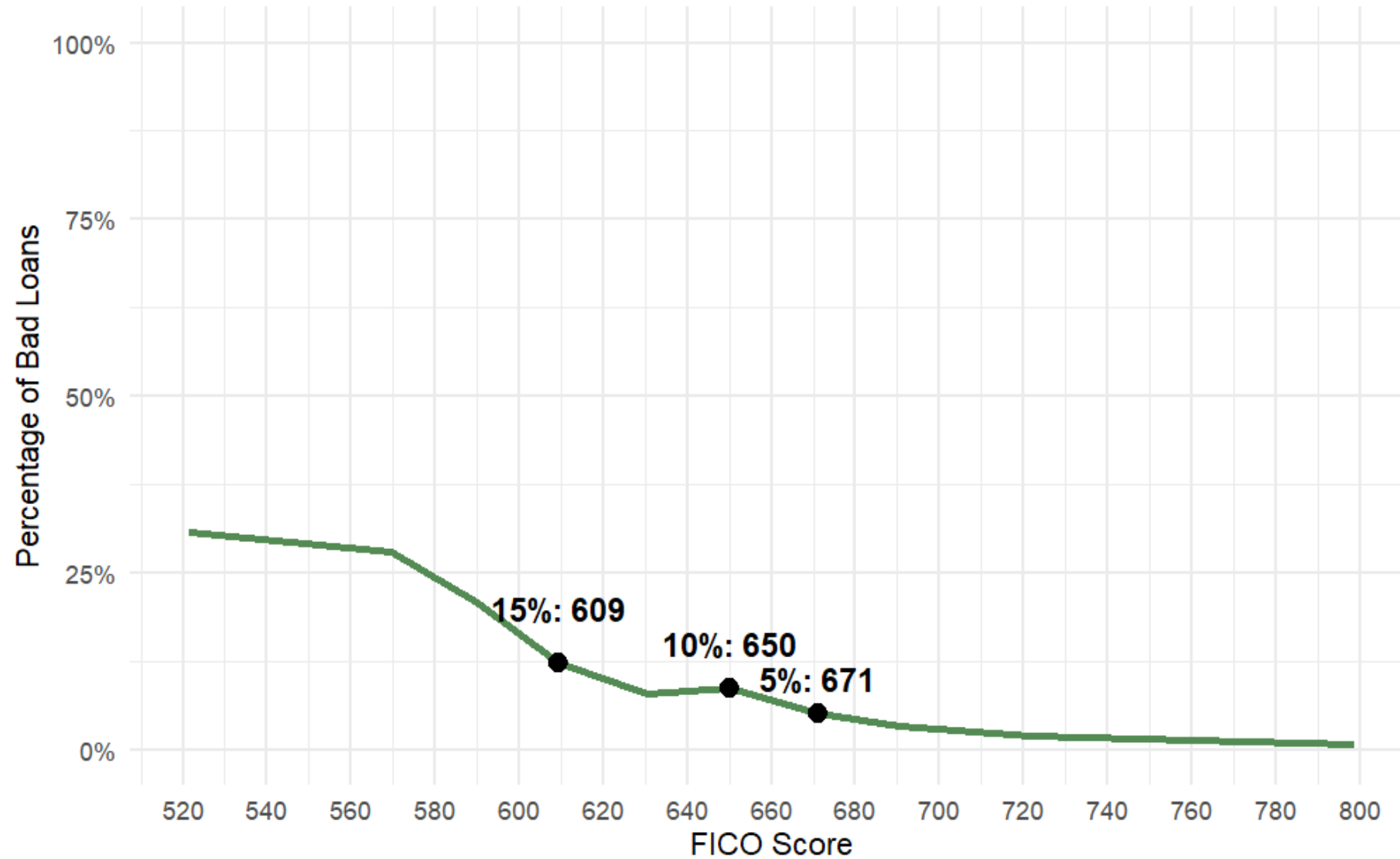
Transaction Year

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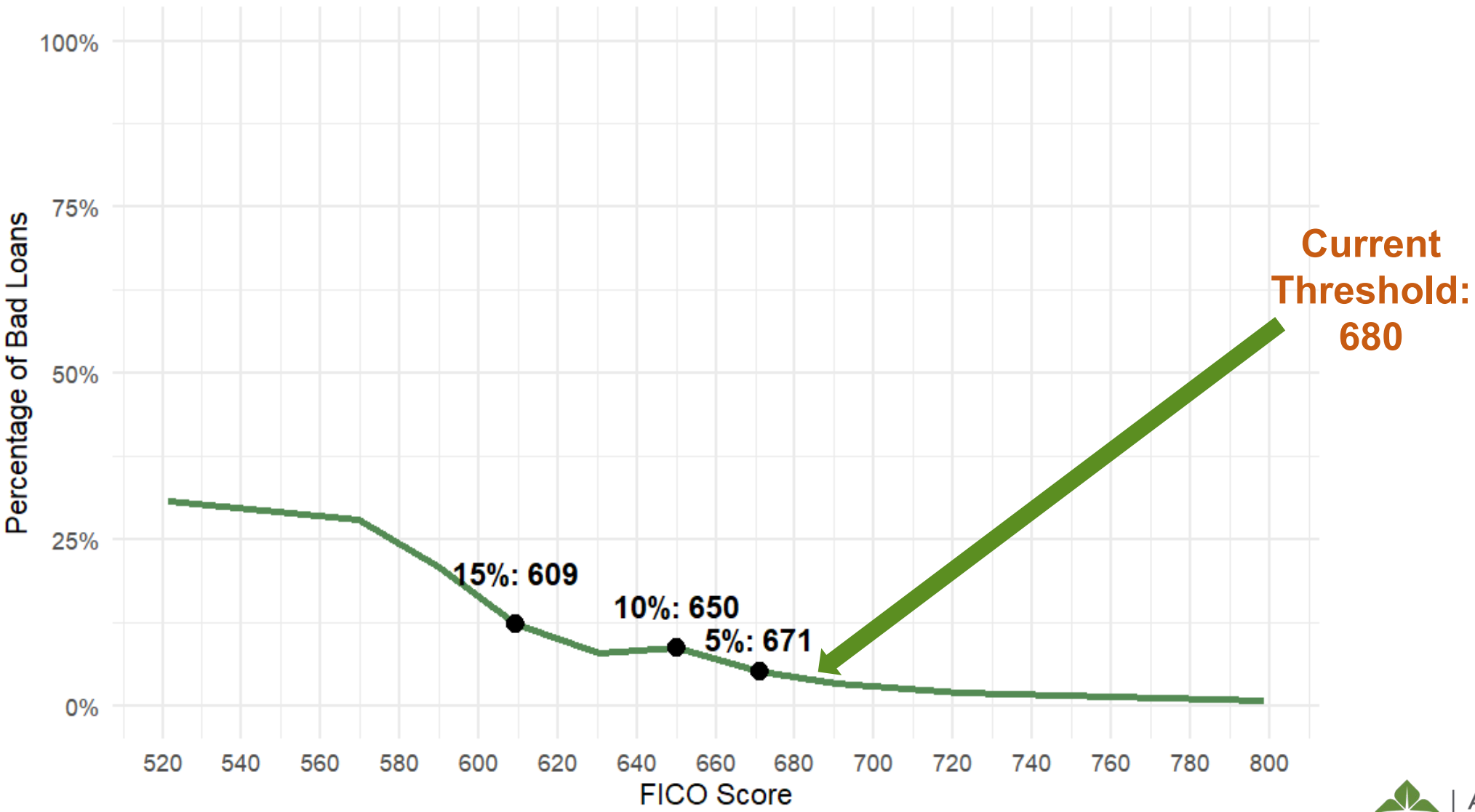
CORRELATION WITH BAD FLAG



FICO AND PERCENT BAD LOANS



FICO AND PERCENT BAD LOANS



FICO TABLE

Fico Scores	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
<600	174	47	127	27.01	20760	0.23
600-604	19	3	16	15.79	20760	0.01
605-609	18	4	14	22.22	20760	0.02
610-614	18	1	17	5.56	20760	0.00
615-619	18	1	17	5.56	20760	0.00
620-624	32	6	26	18.75	20760	0.03
625-629	47	4	43	8.51	20760	0.02
630-634	40	2	38	5.00	20760	0.01
635-639	57	2	55	3.51	20760	0.01
640-644	49	1	48	2.04	20760	0.00
645-649	74	9	65	12.16	20760	0.04
650-654	62	5	57	8.06	20760	0.02
655-659	81	8	73	9.88	20760	0.04
660-664	88	4	84	4.55	20760	0.02
665-669	83	3	80	3.61	20760	0.01
670-674	157	12	145	7.64	20760	0.06
675-679	173	6	167	3.47	20760	0.03
680-684	140	3	137	2.14	20760	0.01
685-689	188	7	181	3.72	20760	0.03
690-694	216	7	209	3.24	20760	0.03
695-700	238	9	229	3.78	20760	0.04
700+	18788	180	18608	0.96	20760	0.87

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FICO WOE TABLE

FICO Scores	Count	Proportions of Obs	WOE	IV
<560	1016	0.00	-3.35	1.43
560-579	270	0.00	-2.84	0.03
580-599	841	0.00	-2.54	0.10
600-619	1356	0.00	-2.34	0.18
620-639	2807	0.01	-1.63	0.24
640-659	4332	0.01	-2.29	0.49
660-679	7541	0.02	-1.31	0.58
680-699	12306	0.04	-0.97	0.64
700-719	17446	0.06	-0.10	0.64
720-739	25409	0.08	-0.55	0.67
740-759	29868	0.10	0.12	0.67
760-780	35357	0.12	1.20	0.77
780+	164322	0.54	1.19	1.22

Weight of Evidence

- **WOE > 0** = proportion of Good loans is higher than Bad (**less risky**)
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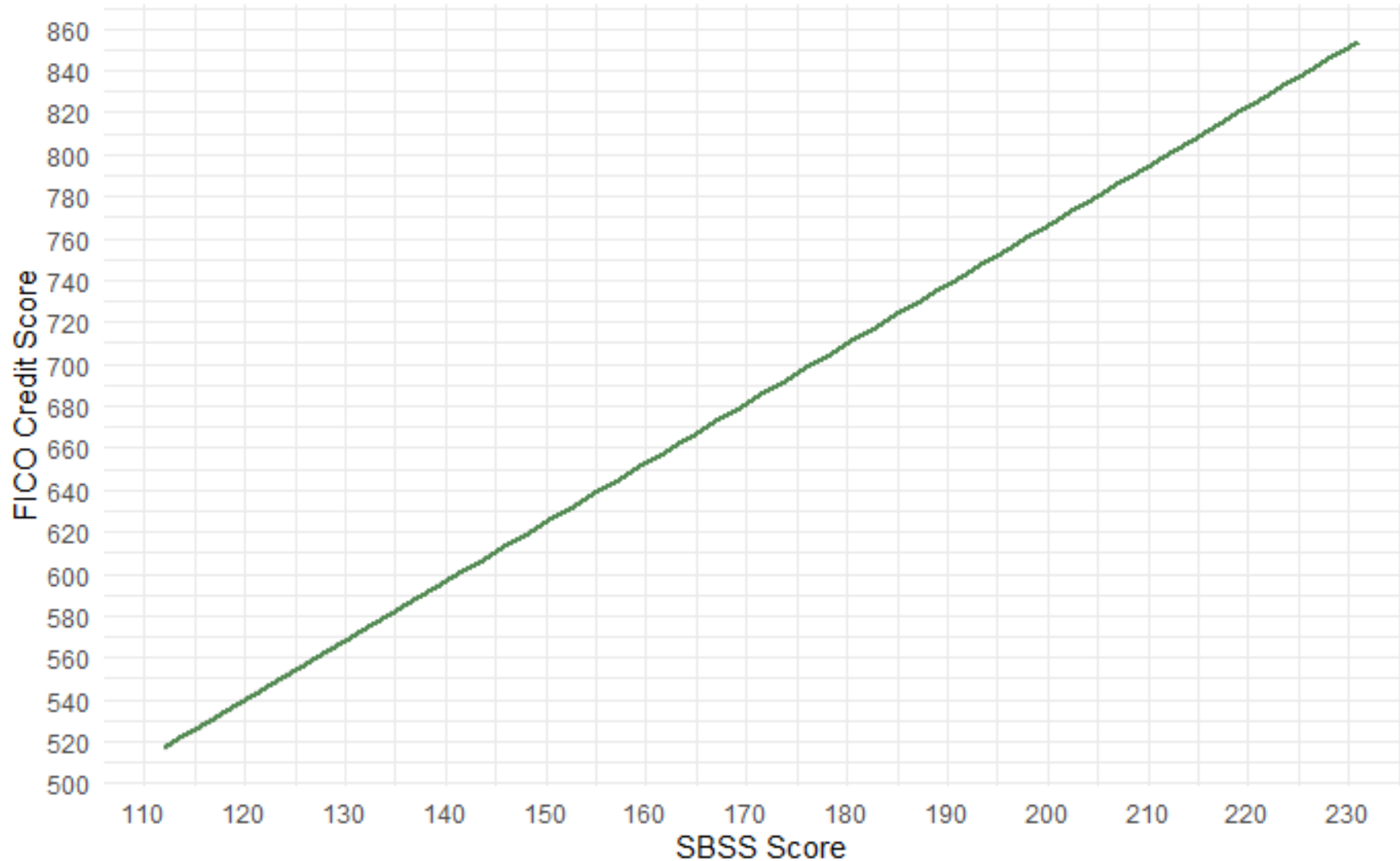
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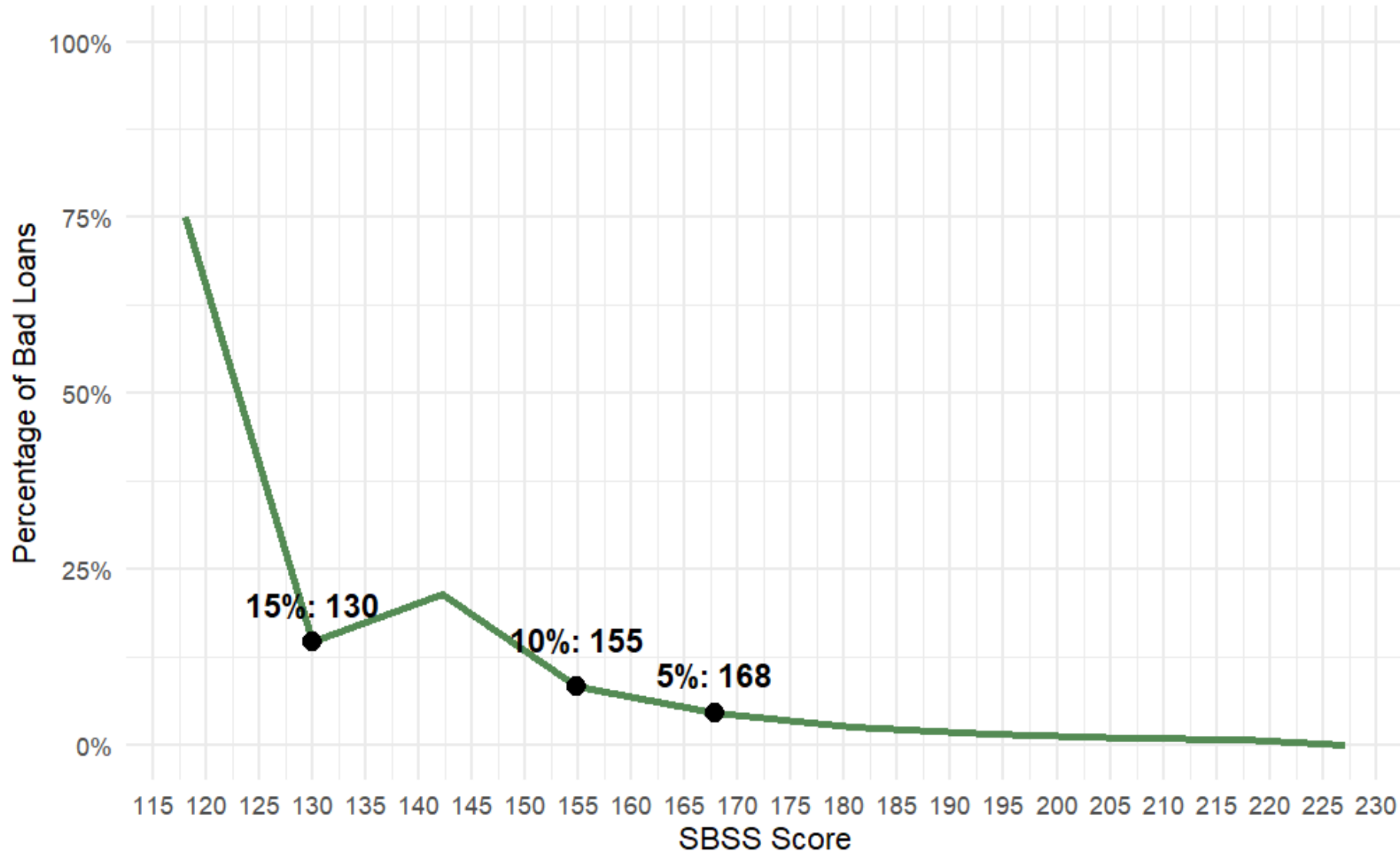
FICO CREDIT SCORE AND SBSS SCORE



Small Business Scoring Service (SBSS score):

- Range: 0-300
- Assesses the credit risk of small businesses applying for loans

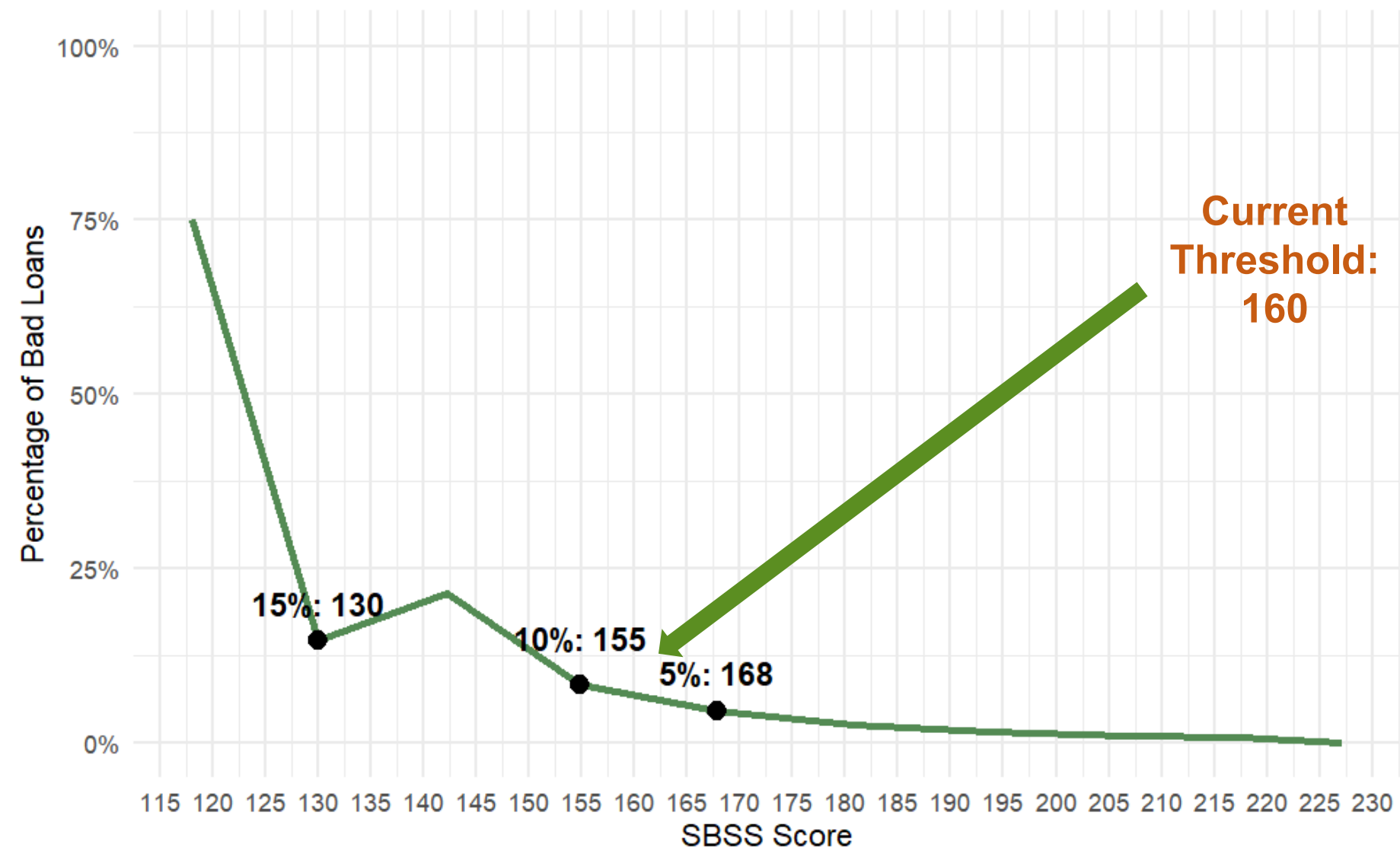
SBSS AND PERCENT BAD LOANS



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SBSS AND PERCENT BAD LOANS



**Current
Threshold:
160**

**Small Business
Scoring Service
(SBSS score):**

- Range: 0-300
- Assesses the credit risk of small businesses applying for loans

SBSS TABLE

SBSS Scores	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
109-114	1	1	0	100.00	20382	0.00
115-119	1	1	0	100.00	20382	0.00
120-124	5	2	3	40.00	20382	0.01
125-129	11	1	10	9.09	20382	0.00
130-134	20	3	17	15.00	20382	0.01
135-139	35	11	24	31.43	20382	0.05
140-144	55	10	45	18.18	20382	0.05
145-149	94	13	81	13.83	20382	0.06
150-154	144	17	127	11.81	20382	0.08
155-159	193	8	185	4.15	20382	0.04
160-164	241	15	226	6.22	20382	0.07
165-169	312	15	297	4.81	20382	0.07
170-174	436	18	418	4.13	20382	0.09
175-179	621	19	602	3.06	20382	0.09
180-184	755	19	736	2.52	20382	0.09
185-189	1076	14	1062	1.30	20382	0.07
190-194	1324	27	1297	2.04	20382	0.13
195-200	1697	21	1676	1.24	20382	0.10
200+	13361	108	13253	0.81	20382	0.53

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122–134	374	0.00	-2.66	0.06
135–147	1978	0.01	-2.88	0.30
148–160	6641	0.02	-1.54	0.41
161–173	13350	0.04	-1.11	0.51
174–186	27757	0.09	-0.80	0.60
187–199	54633	0.18	-0.08	0.60
200–212	90267	0.30	0.60	0.68
213–225	106556	0.35	1.22	0.99
226+	1249	0.00	0.00	0.99

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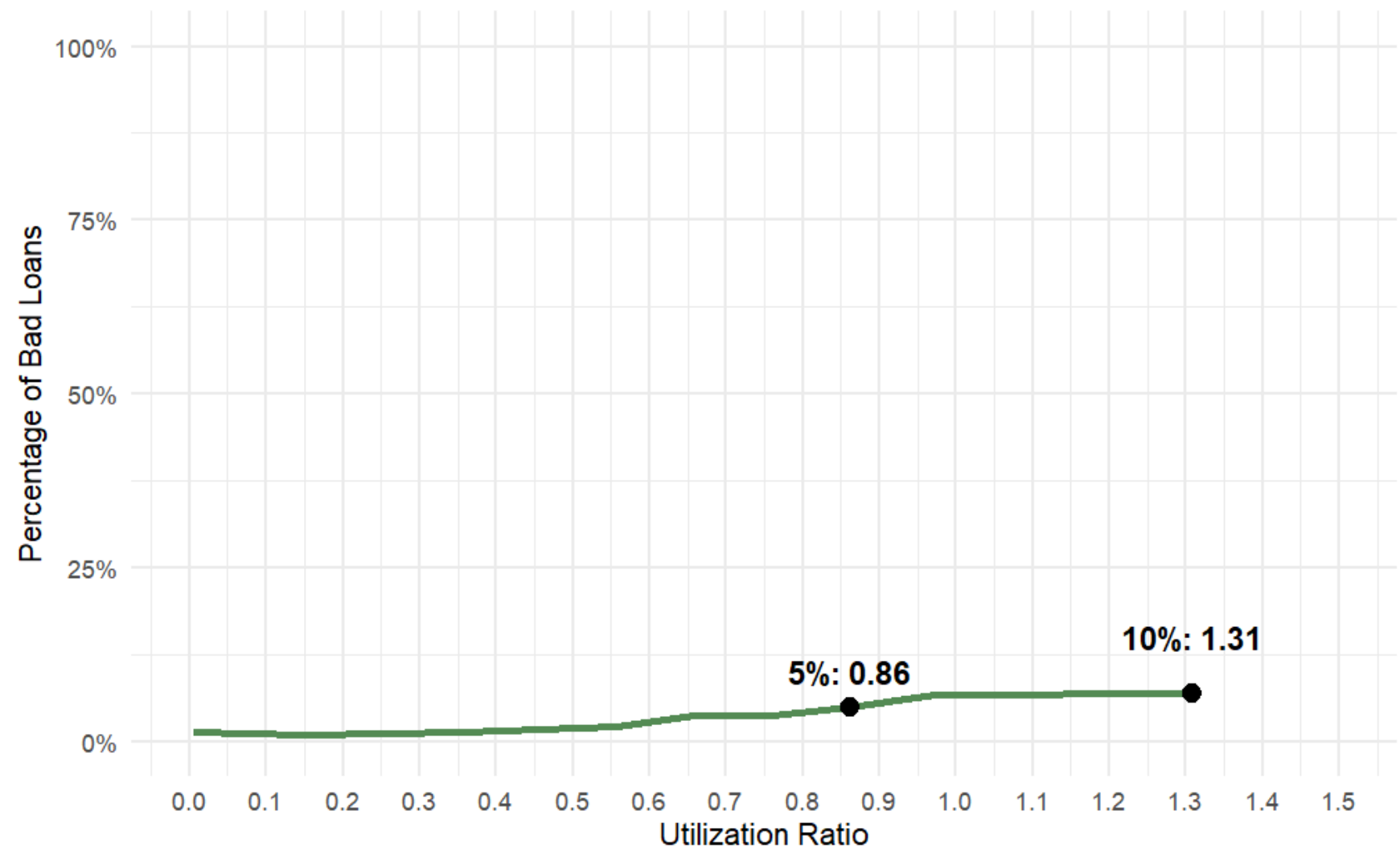
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UTILIZATION RATIO



Utilization Ratio =
Credit Balance
Credit Limit

UTILIZATION RATIO TABLE

Utilization Ratio	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
<.1	37891	481	37410	1.27	70091	0.69
.1-.2	8726	72	8654	0.83	70091	0.10
.21-.3	5046	45	5001	0.89	70091	0.06
.31-.4	3338	36	3302	1.08	70091	0.05
.41-.5	2573	44	2529	1.71	70091	0.06
.51-.6	2267	43	2224	1.90	70091	0.06
.61-.7	2147	75	2072	3.49	70091	0.11
.71-.8	1942	66	1876	3.40	70091	0.09
.81-.9	2253	98	2155	4.35	70091	0.14
.91-1	3682	235	3447	6.38	70091	0.34
1+	226	15	211	6.64	70091	0.02

Utilization Ratio =

Credit Balance

Credit Limit

UTILIZATION RATIO WOE TABLE

Utilization Ratio	Count	Proportions of Obs	WOE	IV
<.1	124795	0.41	0.32	0.13
.1-.2	8829	0.03	0.79	0.01
.21-.3	5090	0.02	0.71	0.02
.31-.4	3359	0.01	0.51	0.02
.41-.5	2600	0.01	0.05	0.02
.51-.6	2290	0.01	-0.05	0.02
.61-.7	2163	0.01	-0.68	0.03
.71-.8	1963	0.01	-0.65	0.03
.81-.9	2265	0.01	-0.91	0.04
.91-1	3691	0.01	-1.33	0.08
1+	145826	0.48	-0.17	0.10

$$\text{Utilization Ratio} = \frac{\text{Credit Balance}}{\text{Credit Limit}}$$

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HIGH UTILIZATION RATIO

High Utilization Flag	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
0	20527	311	20216	1.52	24353	1.28
1	3826	147	3679	3.84	24353	0.60

High Utilization Flag:

- 0 = less than 80% credit usage
- 1 = 80% credit usage or more

HIGH UTILIZATION RATIO

High Utilization Flag	Count	Proportions of Obs	WOE	IV
0	150886	0.96	0.13	0.01
1	6386	0.04	-1.37	0.17

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VARIABLE IMPORTANCE INTERACTIONS

**FICO X High
Utilization Flag**

**SBSS Score X
Utilization Ratio**

**SBSS Score X
Transaction Year**

**FICO Score X
Transaction Year**

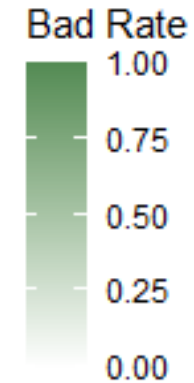
**FICO X
Utilization
Ratio**

**SBSS Score
X High
Utilization
Flag**

FICO AND UTILIZATION RATIO

Bad Loans by FICO Score Bin and Utilization Ratio

FICO Score Bin	Utilization Ratio					
	<.1	.1-.25	.26-.5	.51-.75	.76-1	1+
700+	0.9%	0.6%	0.8%	1.2%	1.7%	0.0%
695-700	4.4%	1.4%	1.0%	6.3%	4.3%	9.1%
690-694	4.9%	5.3%	4.7%	6.1%	4.3%	14.3%
685-689	4.0%	3.9%	1.9%	3.8%	4.5%	0.0%
680-684	1.8%	0.0%	2.5%	2.4%	3.9%	0.0%
675-679	3.5%	3.9%	3.8%	6.2%	4.3%	0.0%
670-674	9.0%	8.0%	4.8%	8.6%	16.0%	0.0%
665-669	6.7%	0.0%	2.3%	2.2%	3.0%	0.0%
660-664	3.5%	2.6%	2.8%	0.0%	3.8%	0.0%
655-659	9.0%	9.1%	7.4%	7.4%	2.8%	0.0%
650-654	8.1%	11.1%	12.1%	12.1%	4.8%	0.0%
645-649	11.1%	23.8%	17.4%	24.4%	15.3%	0.0%
640-644	1.8%	0.0%	5.0%	6.1%	3.8%	66.7%
635-639	2.8%	5.6%	4.2%	6.1%	4.0%	0.0%
630-634	6.4%	0.0%	22.2%	10.0%	8.2%	0.0%
625-629	7.4%	0.0%	13.3%	3.6%	16.7%	0.0%
620-624	23.8%	12.5%	30.8%	20.0%	21.0%	0.0%
615-619	9.5%	14.3%	30.0%	42.9%	10.3%	0.0%
610-614	5.6%	20.0%	0.0%	16.7%	3.1%	0.0%
605-609	19.0%	0.0%	11.1%	17.6%	33.3%	100.0%
600-604	19.0%	0.0%	25.0%	33.3%	16.7%	0.0%
<600	22.6%	25.0%	3.6%	21.0%	31.4%	36.0%

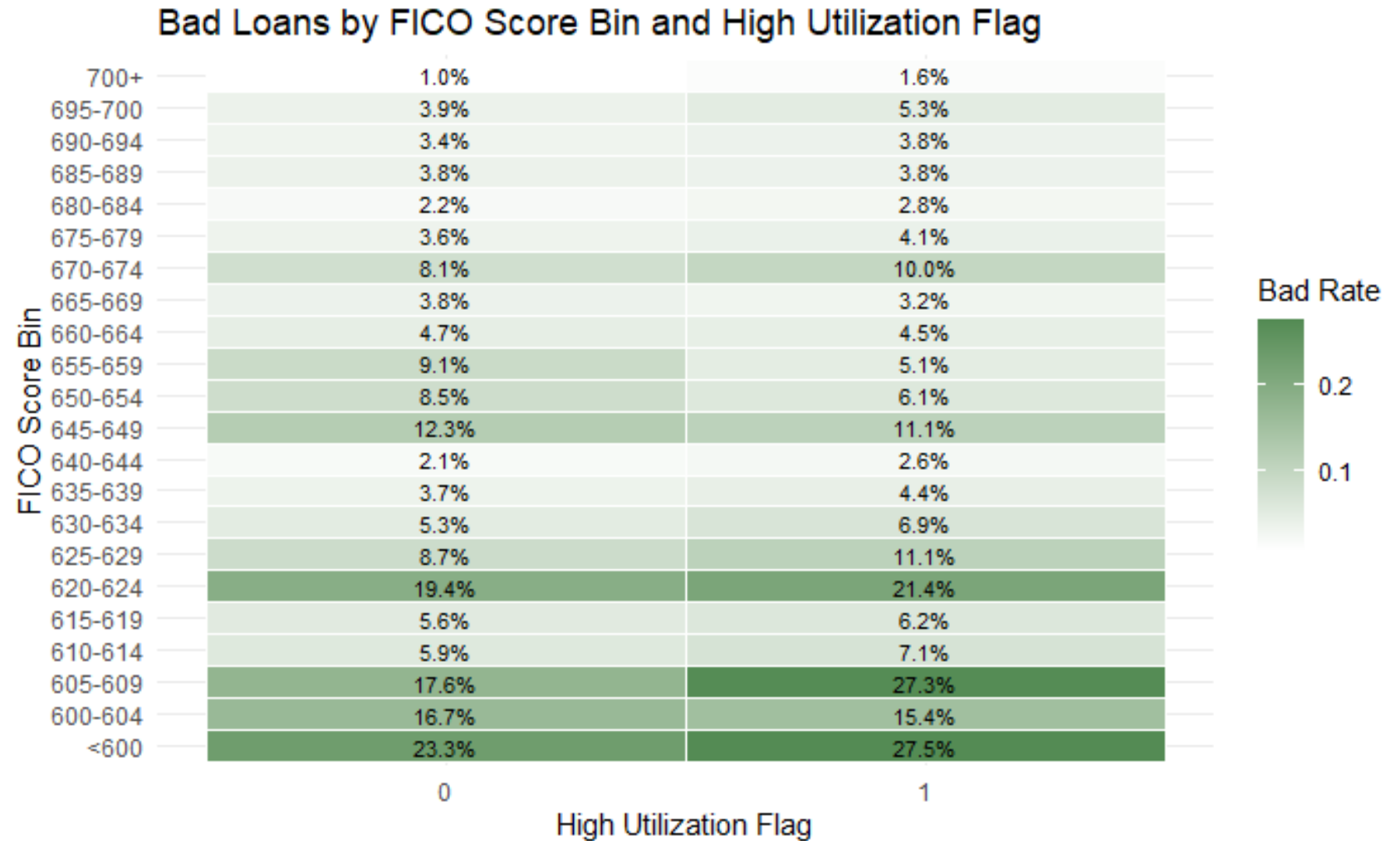


$$\text{Utilization Ratio} = \frac{\text{Credit Balance}}{\text{Credit Limit}}$$

FICO AND HIGH UTILIZATION FLAG

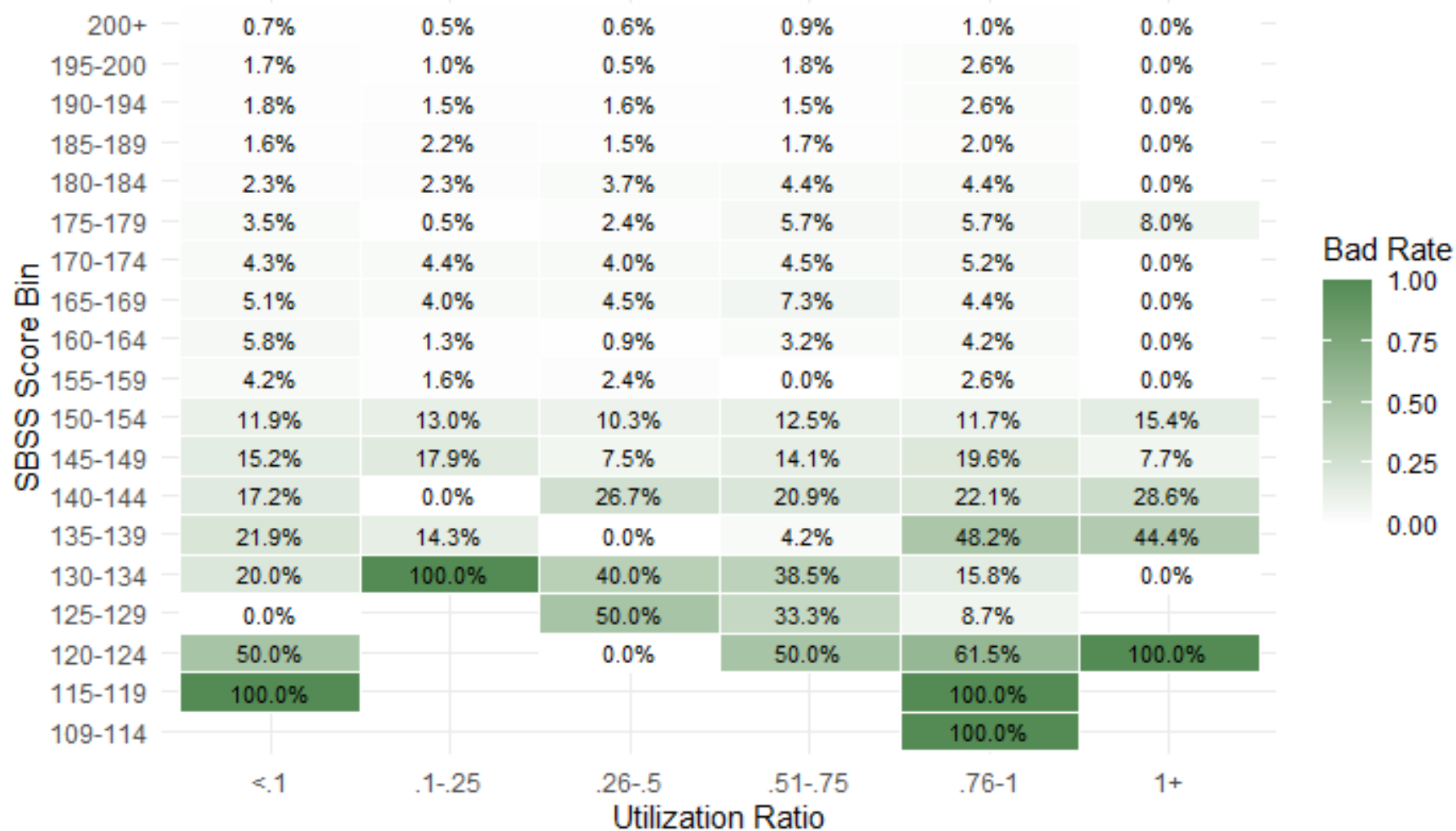
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SBSS AND UTILIZATION RATIO

Bad Loans by SBSS Score Bin and Utilization Ratio

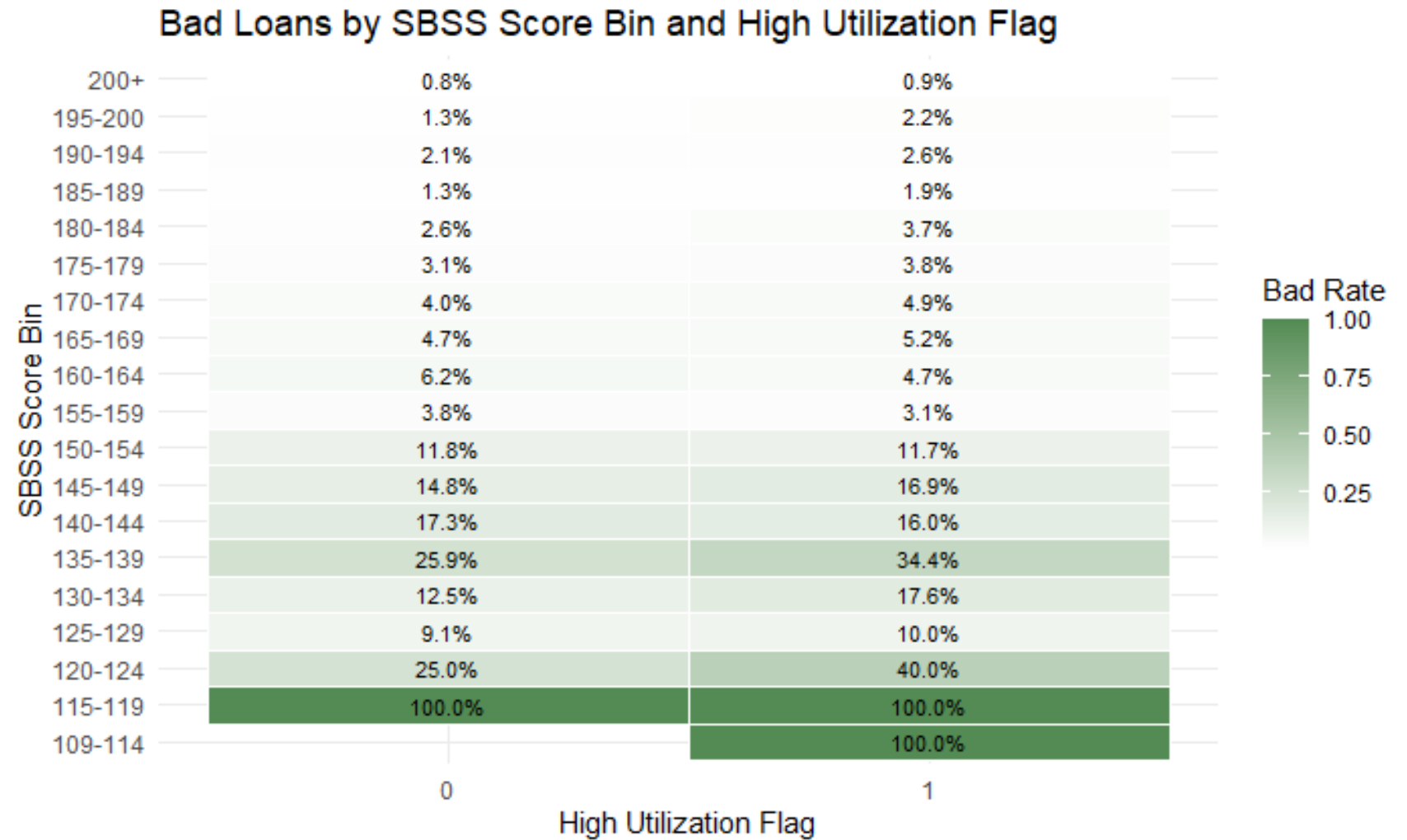


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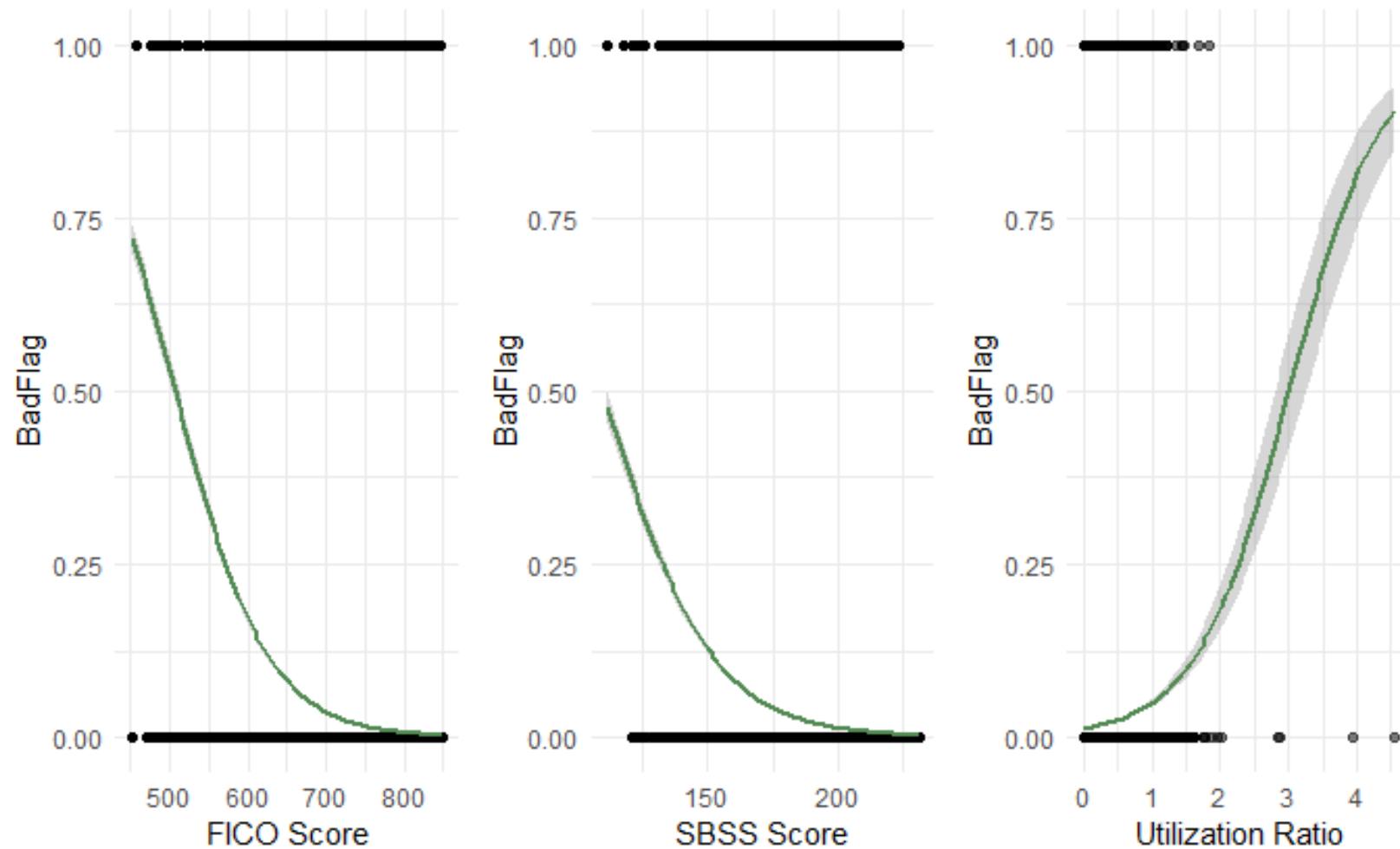


MODEL RECOMMENDATION

Model Type	Model Output	Interpretable	Predictive Power	Transparent + Explainable	Handles Non-Linearity
Logistic Regression	Probability Score (Range: 0-1)	Very	Moderately	Yes	No
Decision Tree	Category Prediction	Moderately	Moderately	Yes	Yes
Random Forest	Category Prediction (based on all decision trees)	Not Very	Very	Sometimes	Yes



IS THE DATA LINEAR?



Yes!

MODEL RECOMMENDATION

Model Type	Model Output	Interpretable	Predictive Power	Transparent + Explainable	Handles Non-Linearity
Logistic Regression	Probability Score (Range: 0-1)	Very	Moderately	Yes	No
Decision Tree	Category Prediction	Moderately	Moderately	Yes	Yes
Random Forest	Category Prediction (based on all decision trees)	Not Very	Very	Sometimes	Yes

BEHAVIORIAL SCORECARD

Logistic Regression Model



Behavioral Scorecard

Example Scorecard

Variable	Range	WOE	Points
Fico Score	< 560	-3.35	10
	560 - 579	-2.84	20
			
SBSS Score	109-121	-3.77	9
	122-134	-2.66	21
			

More points = More eligible to auto-revolve

THANK YOU AAC FOR THE SUPPORT!

Data Science Team

Minet Polsinelli

Julia Gyamfi

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Jessica Scheer

Brandon Hatter

IT Support

Derrick May

Zielke Pham

THANK YOU!

Any Questions?



American
AgCredit

APPENDIX



American
AgCredit





VARIABLE IMPORTANCE (REVOLVER, TU AND CREATED VARIABLES)

	Variable <chr>	IV <dbl>
33	LineUsage	3.014917866248
34	LineUsage12moAvg	2.145457105945
29	fico9score	1.281562526736
30	sbss7score	0.986022227579
38	num_revolver_flag_accounts	0.631550232354
46	accounts_last_12_months	0.594000632186
40	num_accounts_softpull	0.528763021006
26	observation_date	0.364661490691
2	transaction_year	0.351661553204
45	accounts_last_6_months	0.267711076695

	Variable <chr>	IV <dbl>
3	transaction_day	0.266826467379
42	avg_account_age_months	0.255529303494
13	terms_payment_schedule_month_count	0.186810592503
4	how_many_times_tr01	0.167895211413
44	avg_account_age_months_TU	0.127479247980
16	account_type	0.123200350480
36	utilization_ratio	0.118700648932
37	high_utilization_flag	0.110449039104
1	record_id	0.084877202775
43	months_open_TU	0.080709210947

VARIABLE IMPORTANCE (REVOLVER, ALL VARIABLES)

	Variable <chr>	IV <dbl>
15	CRGFlag	3.769540556033
12	DaysPastDue	3.103427986776
52	LineUsage	3.014917866248
53	LineUsage12moAvg	2.145457105945
13	TimesLate60DaysLTD	1.716321194307
9	LoanLevelPD_Rating	1.709740421498
48	fico9score	1.281562526736
1	months_open	1.083511951935
49	sbss7score	0.986022227579
6	ImpairFlag	0.723832792875

	Variable <chr>	IV <dbl>
8	NonAccrualFlag	0.714786591116
57	num_revolver_flag_accounts	0.631550232354
65	accounts_last_12_months	0.594000632186
59	num_accounts_softpull	0.528763021006
10	ApprovedAmtAtOrigination	0.414796470741
45	observation_date	0.364661490691
21	transaction_year	0.351661553204
2	EnterpriseCustomerID	0.324453966324
11	RelatedLnAttributionGrossAmt	0.283193029420
64	accounts_last_6_months	0.267711076695

	Variable <chr>	IV <dbl>
22	transaction_day	0.266826467379
61	avg_account_age_months	0.255529303494
32	terms_payment_schedule_month_count	0.186810592503
14	NetChargeOffAmtLTD	0.171797263198
19	ChargeOffFlag	0.171797263198
23	how_many_times_tr01	0.167895211413
63	avg_account_age_months_TU	0.126336596333
35	account_type	0.123200350480
55	utilization_ratio	0.118700648932
56	high_utilization_flag	0.110449039104

VARIABLE IMPORTANCE (ALL LOANS, ALL VARIABLES)

15	CRGFlag	3.65332653018
9	LoanLevelPD_Rating	2.31772321294
12	DaysPastDue	2.12990715397
49	fico9score	1.42238669210
50	sbss7score	1.15839503998
1	months_open	1.11698946267
13	TimesLate60DaysLTD	1.08698918426
6	ImpairFlag	0.99846156429
8	NonAccrualFlag	0.98645903670
46	observation_date	0.58071069804

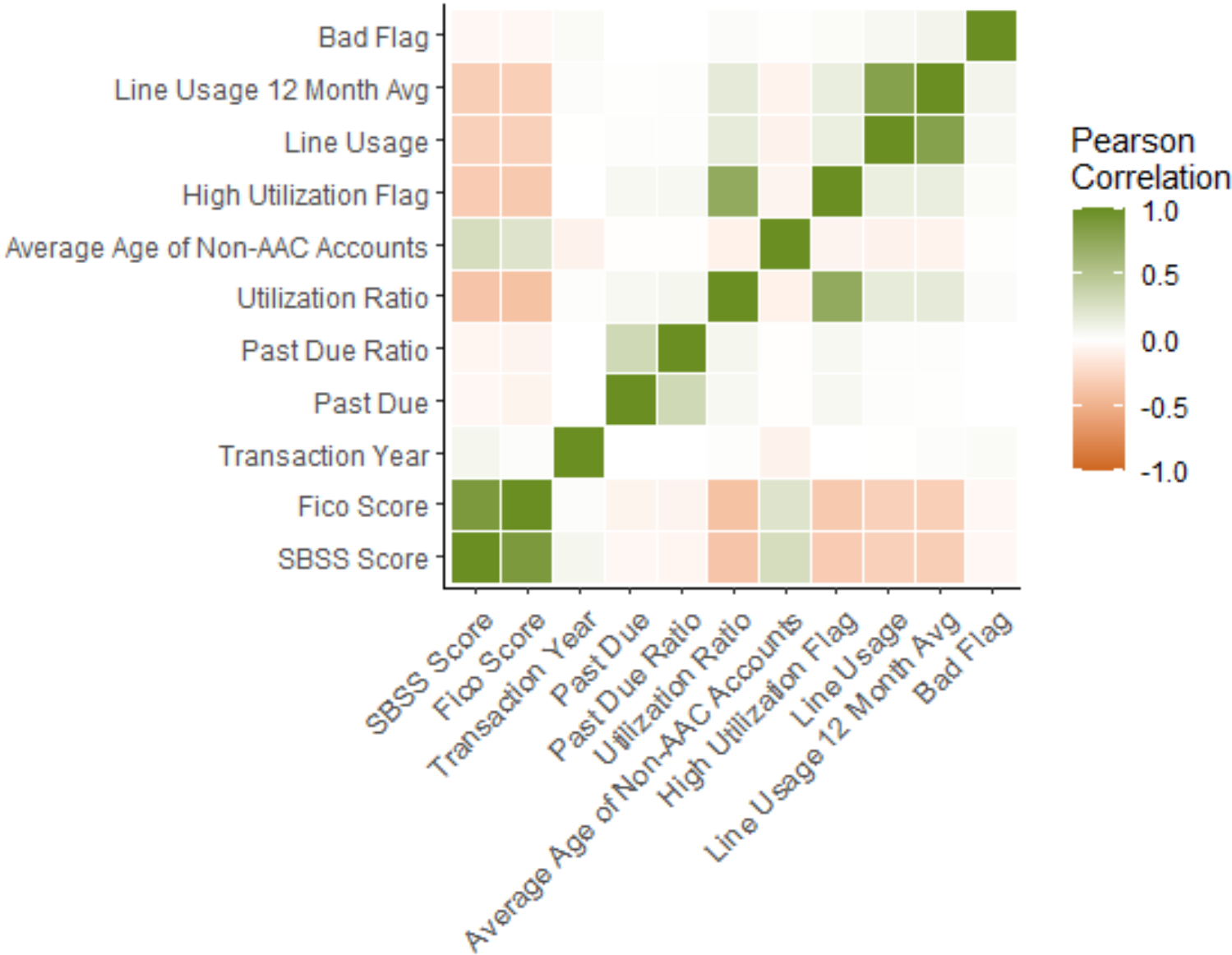
21	transaction_year	0.55970664739
22	transaction_day	0.47484209868
54	LineUsage12moAvg	0.46205625309
4	LoanTypeDesc	0.39097725756
3	LoanCategory	0.38522062899
59	num_revolver_flag_accounts	0.26224099173
53	LineUsage	0.23407237282
60	num_AAC_loans	0.19017819143
63	avg_account_age_months	0.17682773969
67	accounts_last_12_months	0.17477916325

2	EnterpriseCustomerID	0.16988584901
66	accounts_last_6_months	0.15953505297
11	RelatedLnAttributionGrossAmt	0.14722531235
47	product_type_num	0.14428807330
48	prod_typ	0.14428807330
61	num_accounts_softpull	0.14398235927
42	times_30_days_late	0.13857401017
32	terms_payment_schedule_month_count	0.12994960519
65	avg_account_age_months_TU	0.12951417862
57	high_utilization_flag	0.12566469205

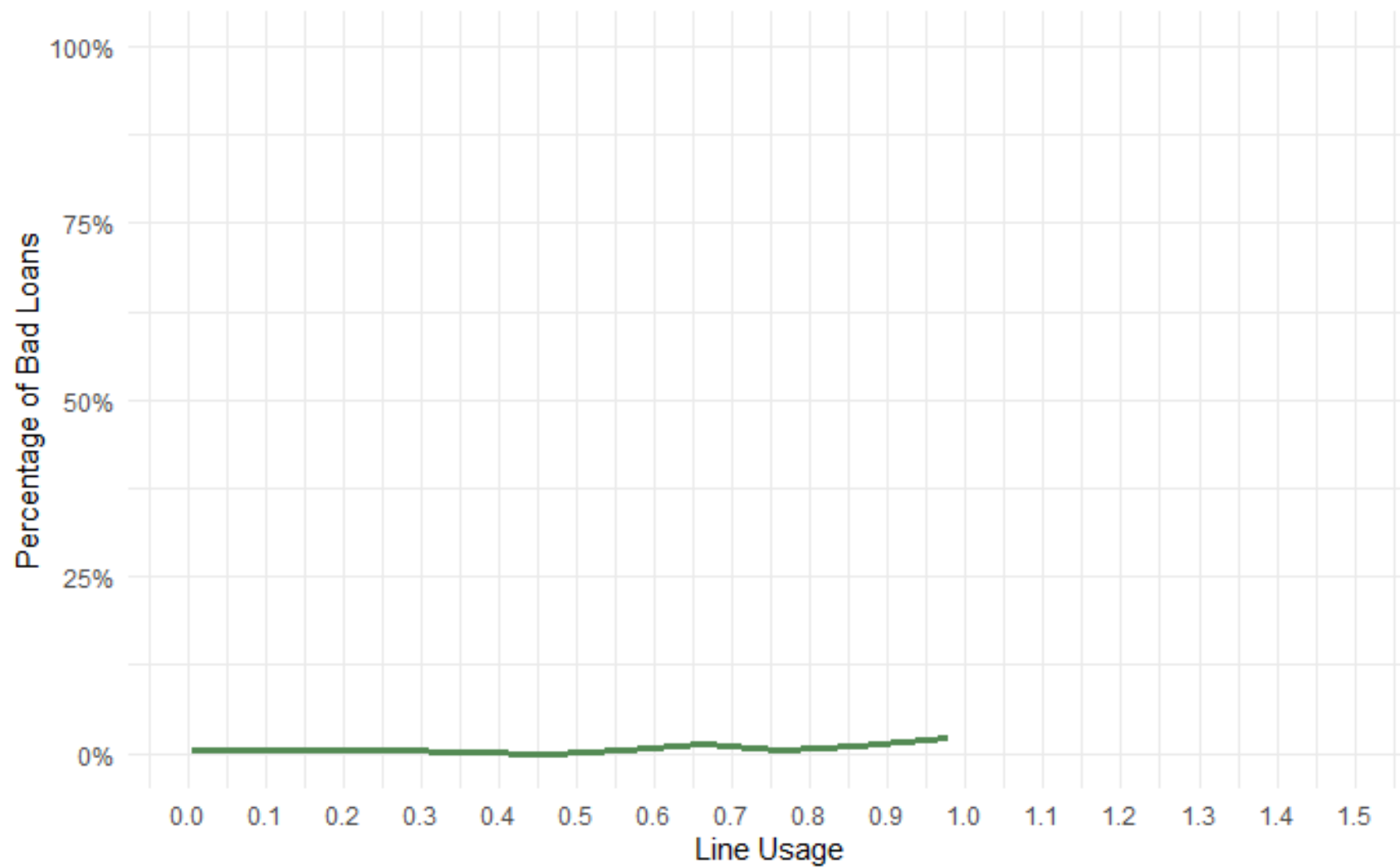
56	utilization_ratio	0.12082311853
36	remark_code	0.11829234507
37	past_due	0.11481563992
39	max_delinquency_date	0.11437445508
14	NetChargeOffAmtLTD	0.11428069727
19	ChargeOffFlag	0.11428069727
10	ApprovedAmtAtOrigination	0.11290494558
18	BusinessCategory	0.10977335378
52	quarter	0.10530550977
43	times_60_days_late	0.10516865693
64	months_open_TU	0.10055816206



CORRELATION WITH BAD FLAG



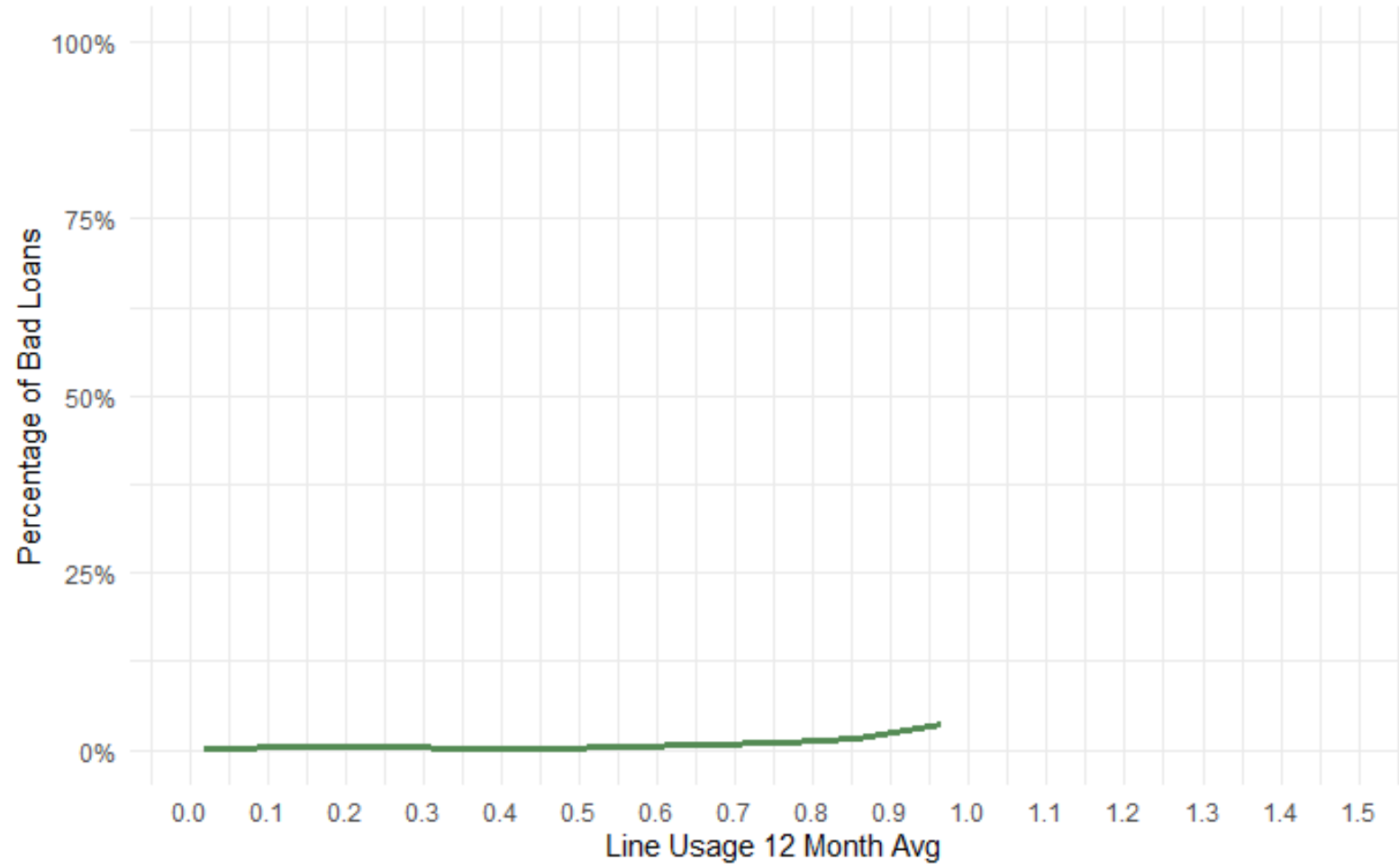
LINE USAGE



LINE USAGE TABLE

Line Usage	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
<.1	17925	41	17884	0.23	55810	0.07
.1-.2	2757	11	2746	0.40	55810	0.02
.21-.3	2575	12	2563	0.47	55810	0.02
.31-.4	2705	5	2700	0.18	55810	0.01
.41-.5	3038	0	3038	0.00	55810	0.00
.51-.6	3334	8	3326	0.24	55810	0.01
.61-.7	3821	52	3769	1.36	55810	0.09
.71-.8	3696	41	3655	1.11	55810	0.07
.81-.9	4567	42	4525	0.92	55810	0.08
.91-1	11392	234	11158	2.05	55810	0.42

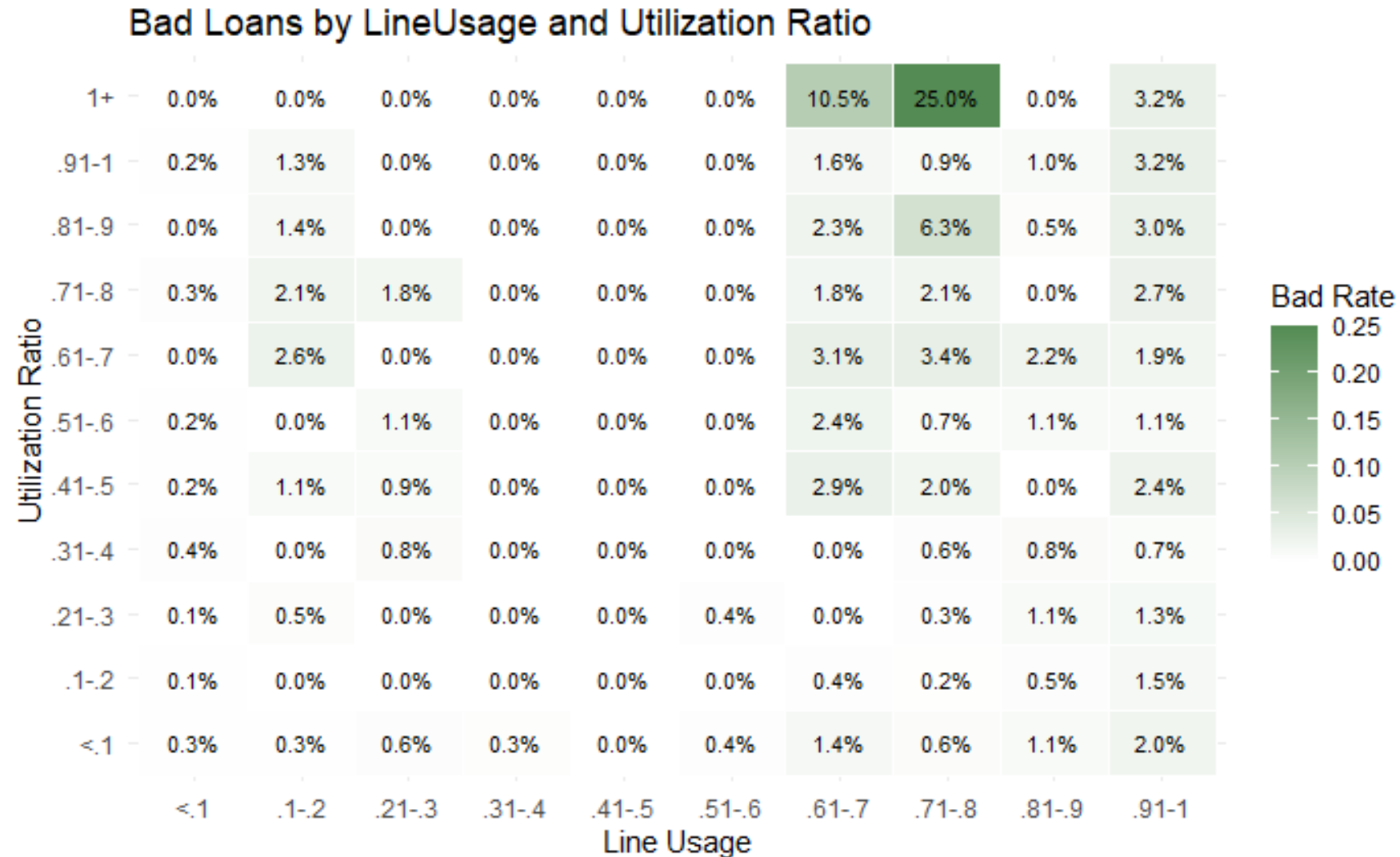
LINE USAGE 12 MONTH AVERAGE



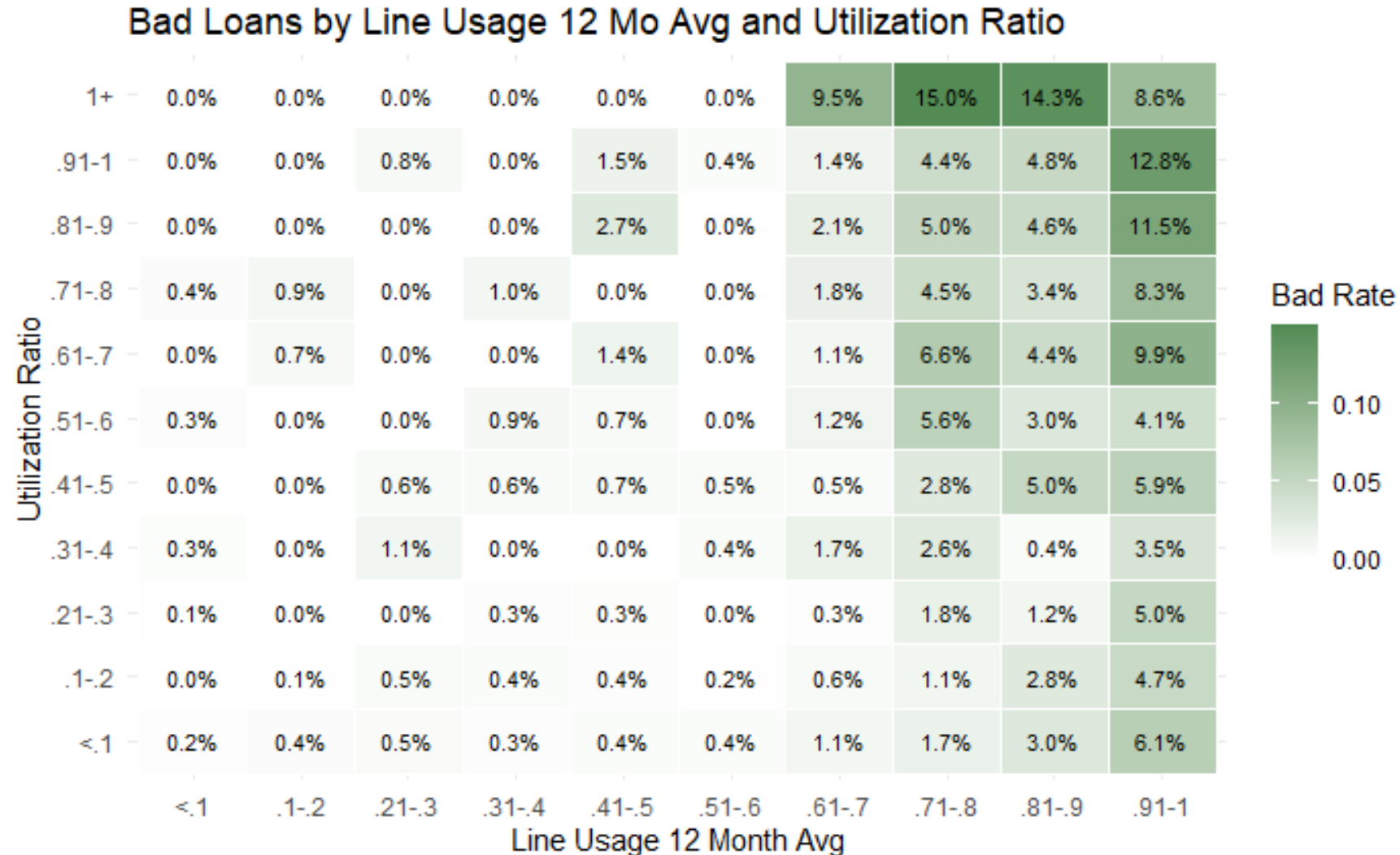
LINE USAGE 12 MONTH AVERAGE TABLE

Line Usage 12 Month Average	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
<.1	14485	26	14459	0.18	57617	0.05
.1-.2	5187	16	5171	0.31	57617	0.03
.21-.3	3888	17	3871	0.44	57617	0.03
.31-.4	3961	13	3948	0.33	57617	0.02
.41-.5	4255	22	4233	0.52	57617	0.04
.51-.6	4701	14	4687	0.30	57617	0.02
.61-.7	4904	53	4851	1.08	57617	0.09
.71-.8	5145	136	5009	2.64	57617	0.24
.81-.9	5050	159	4891	3.15	57617	0.28
.91-1	6041	441	5600	7.30	57617	0.77

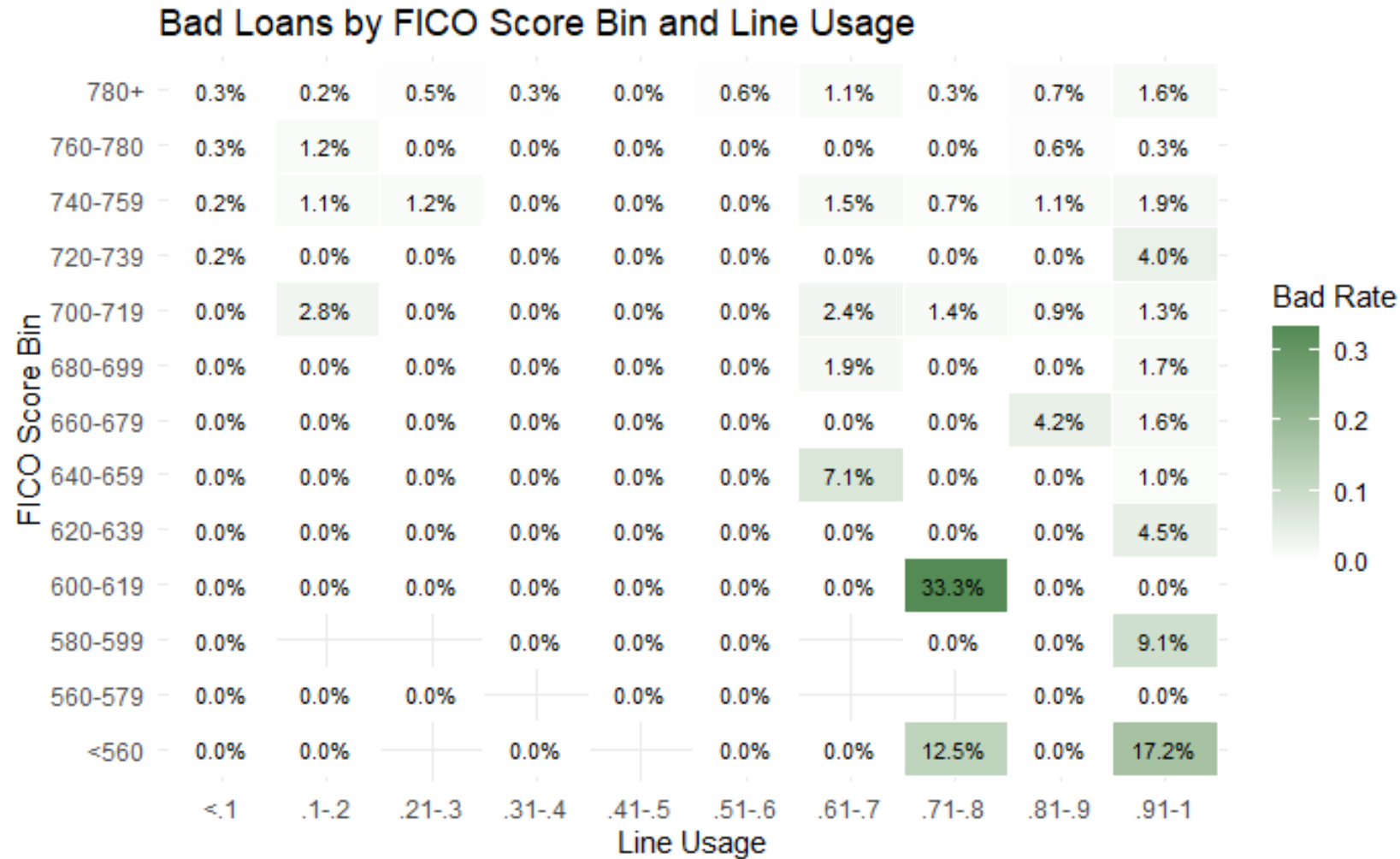
UTILIZATION RATIO AND LINE USAGE



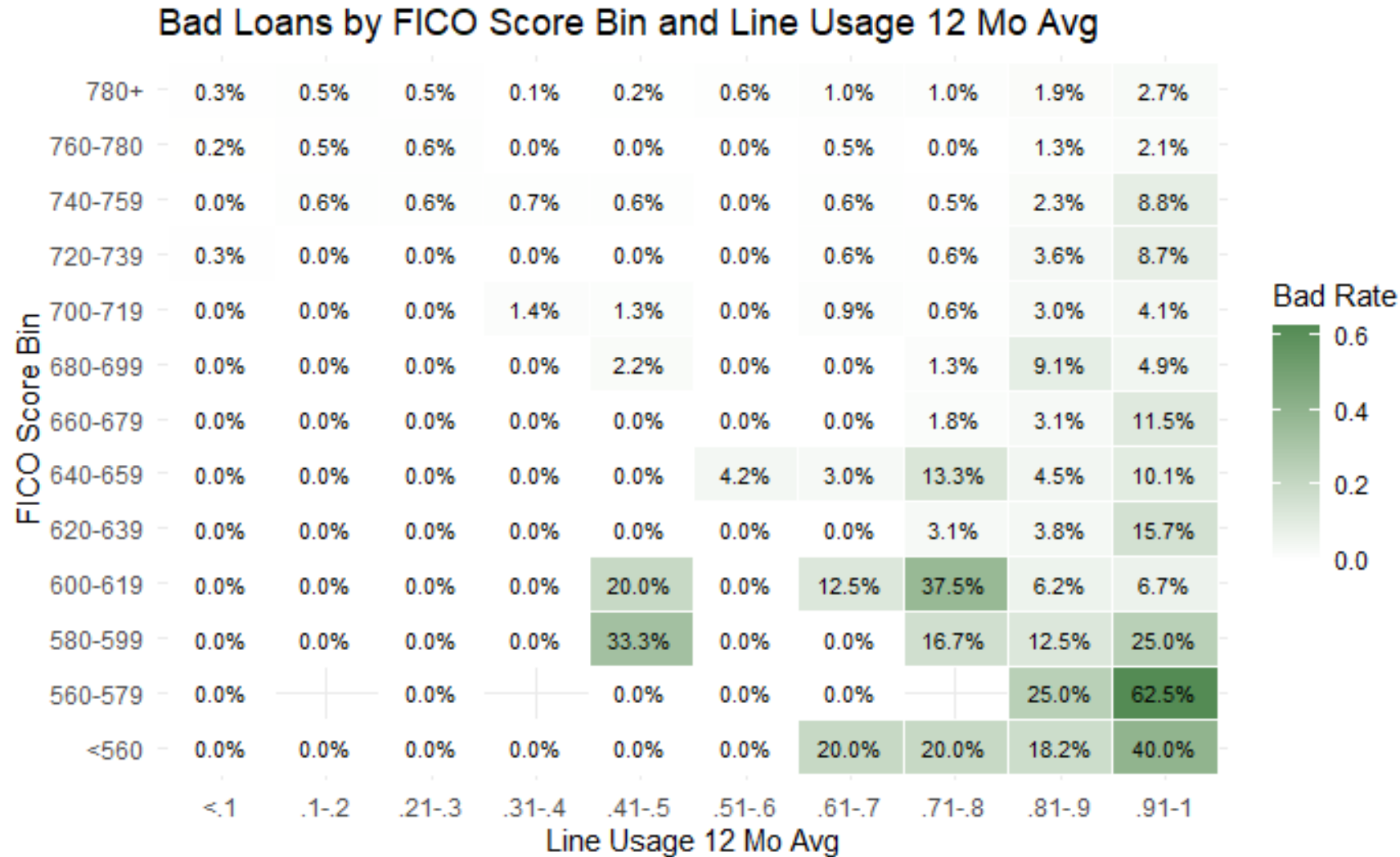
UTILIZATION RATIO AND LINE USAGE 12 MO AVG



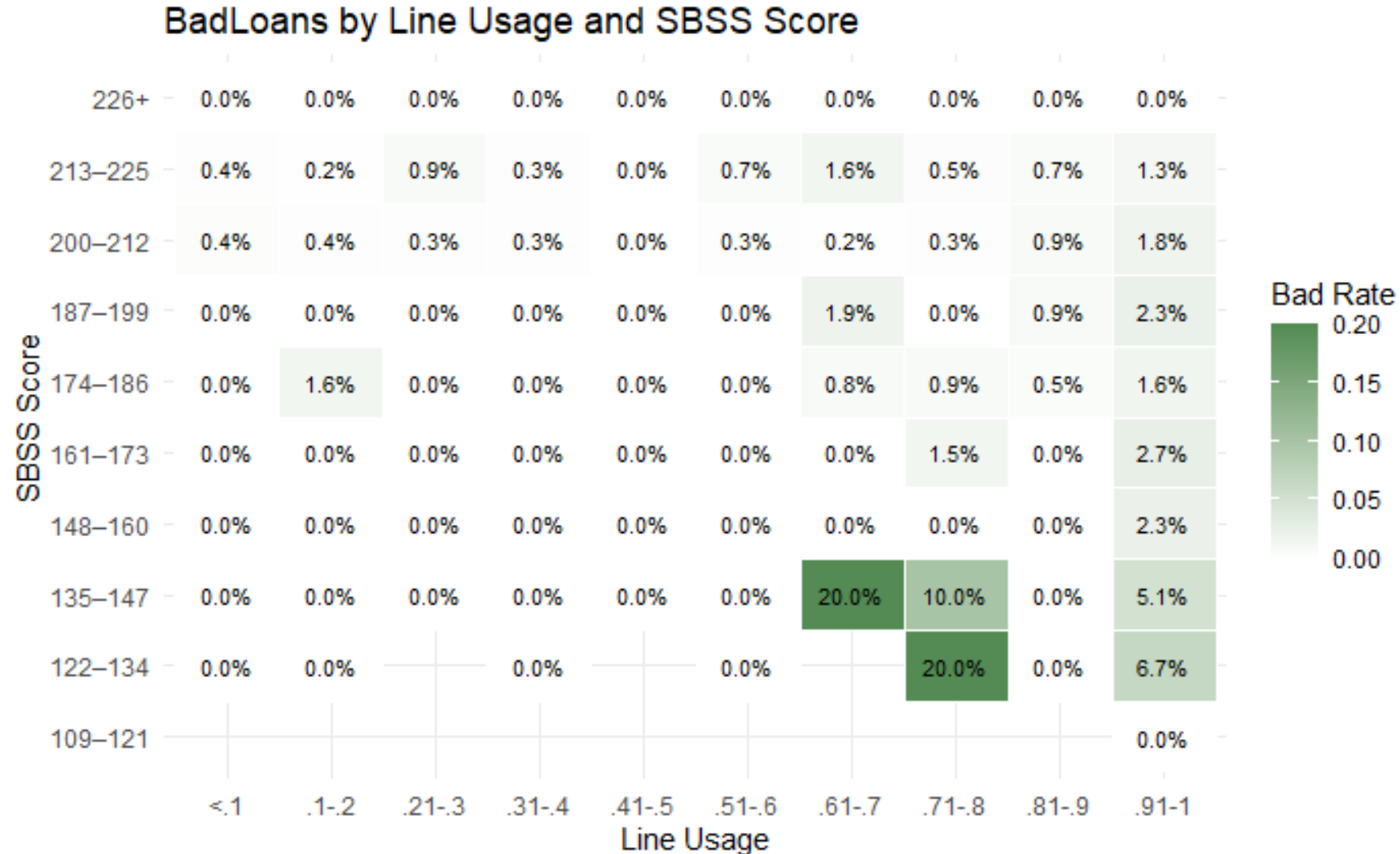
FICO AND LINE USAGE



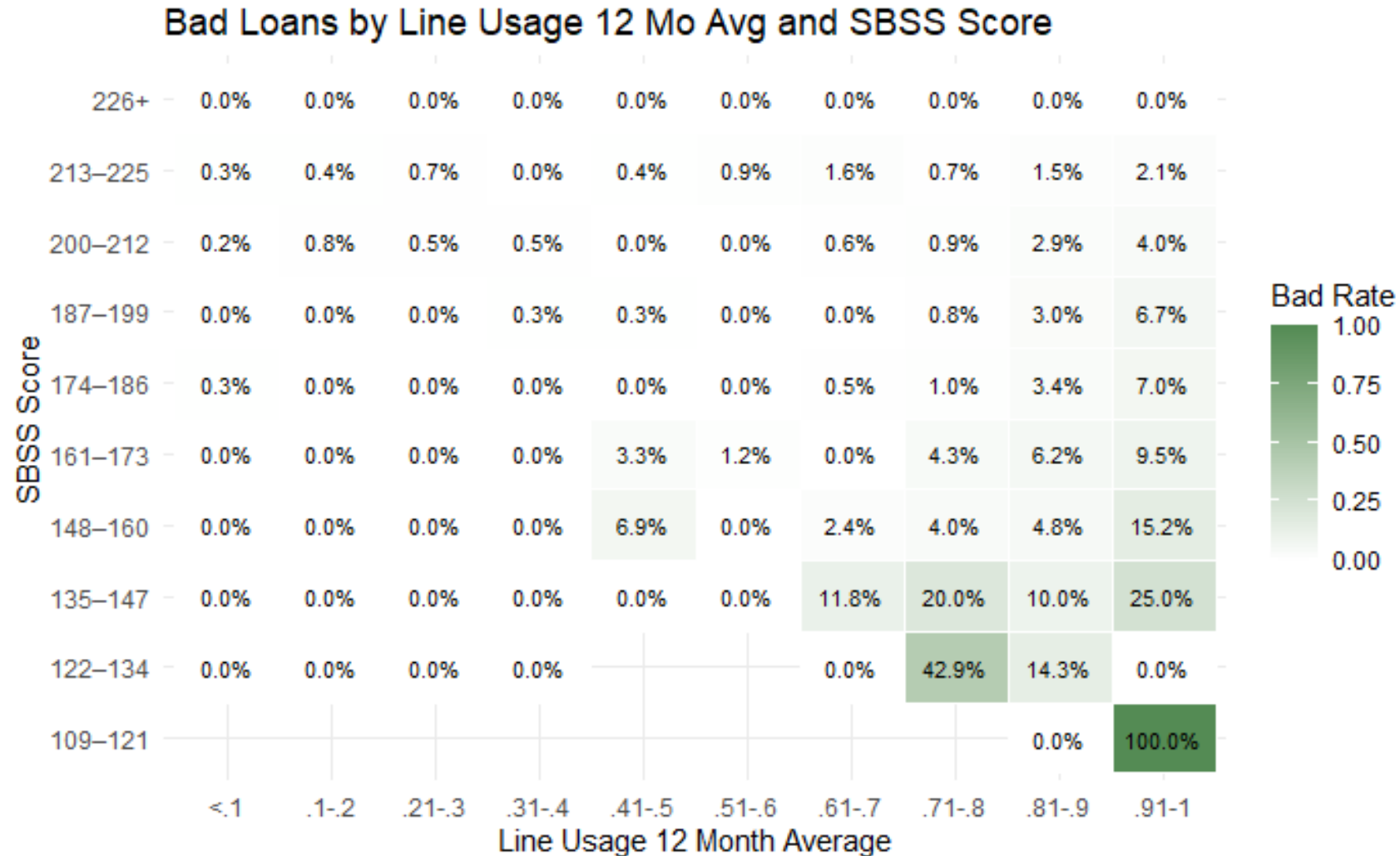
FICO AND LINE USAGE 12 MO AVG



SBSS AND LINE USAGE

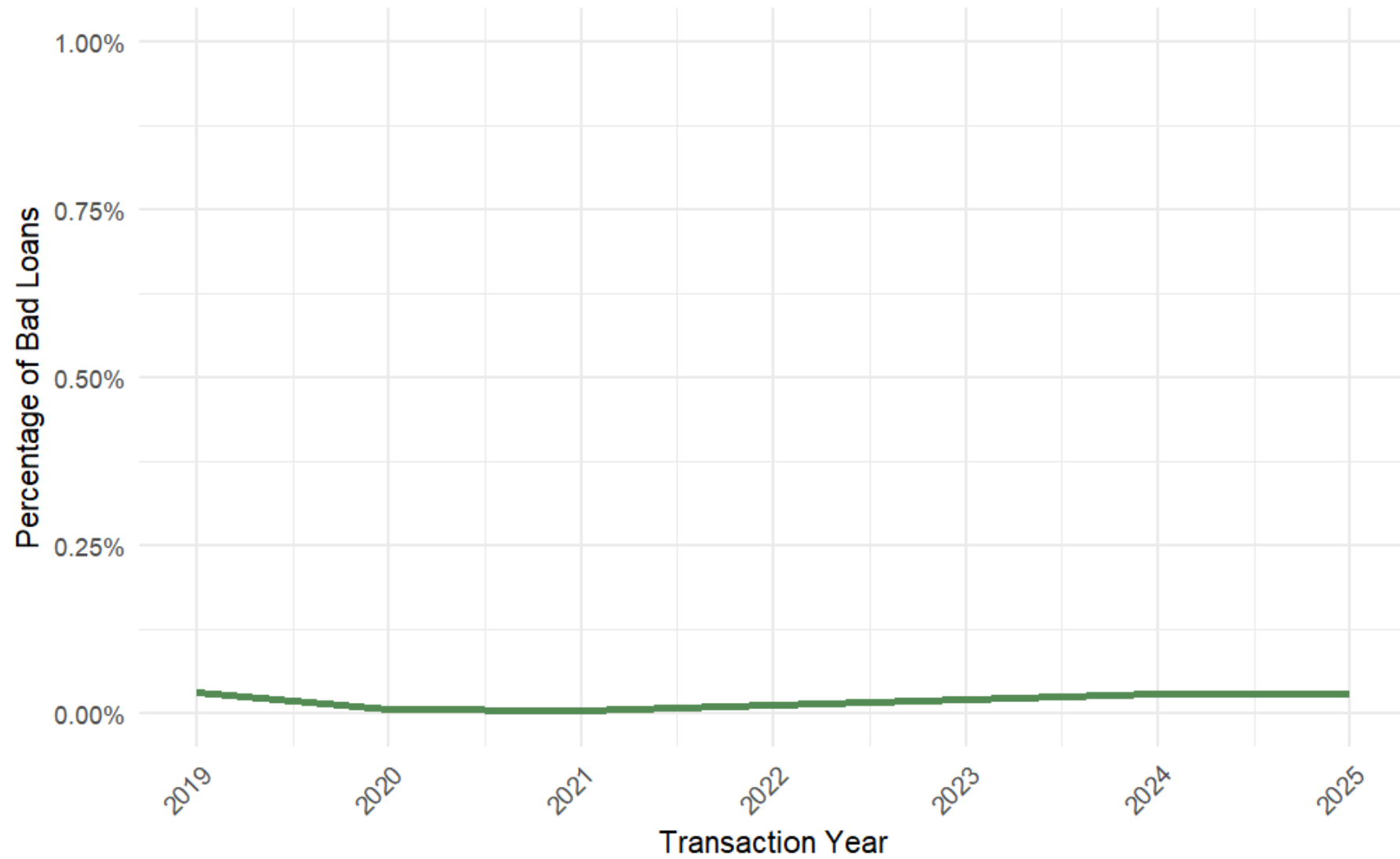


SBSS AND LINE USAGE 12 MO AVG

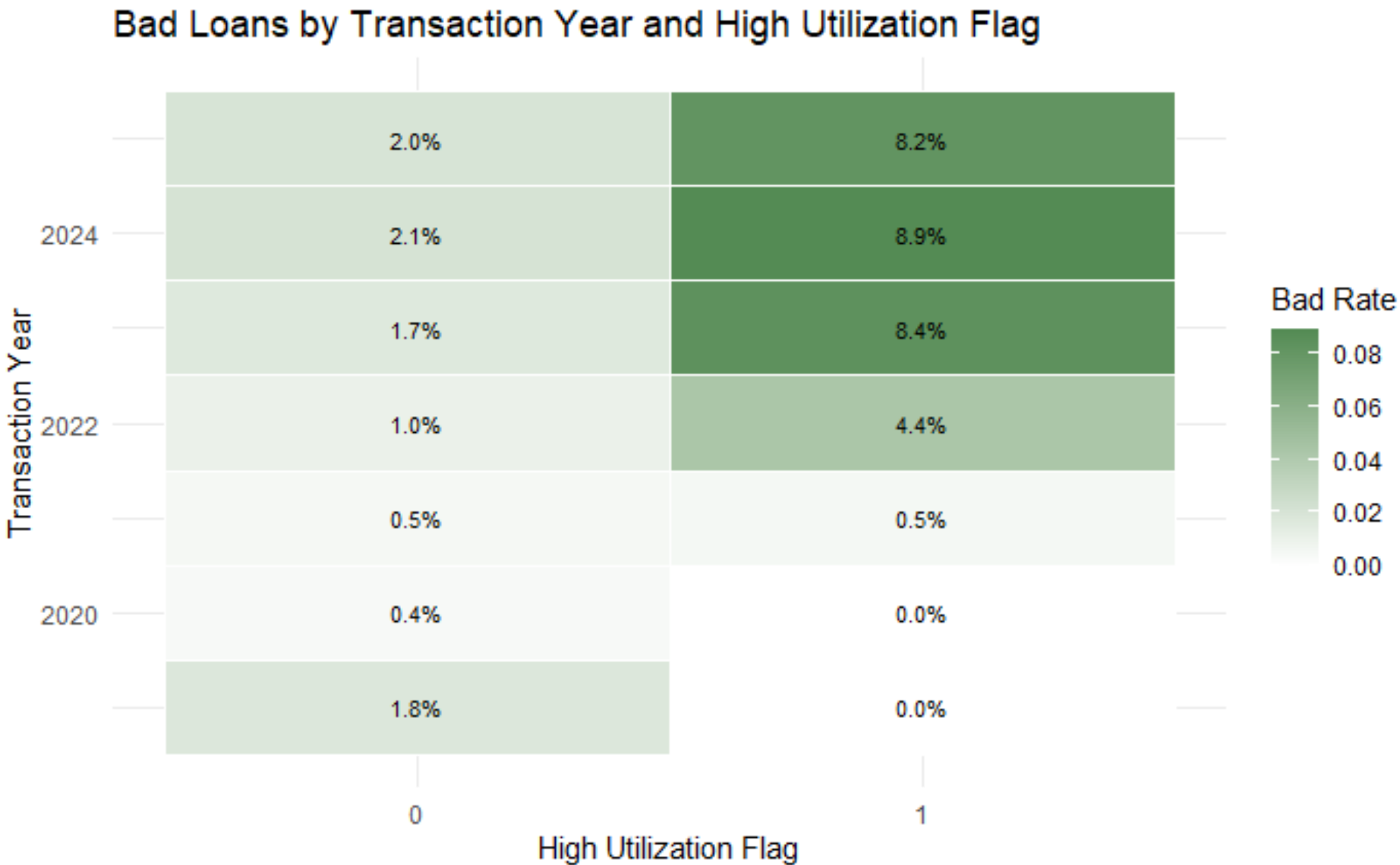




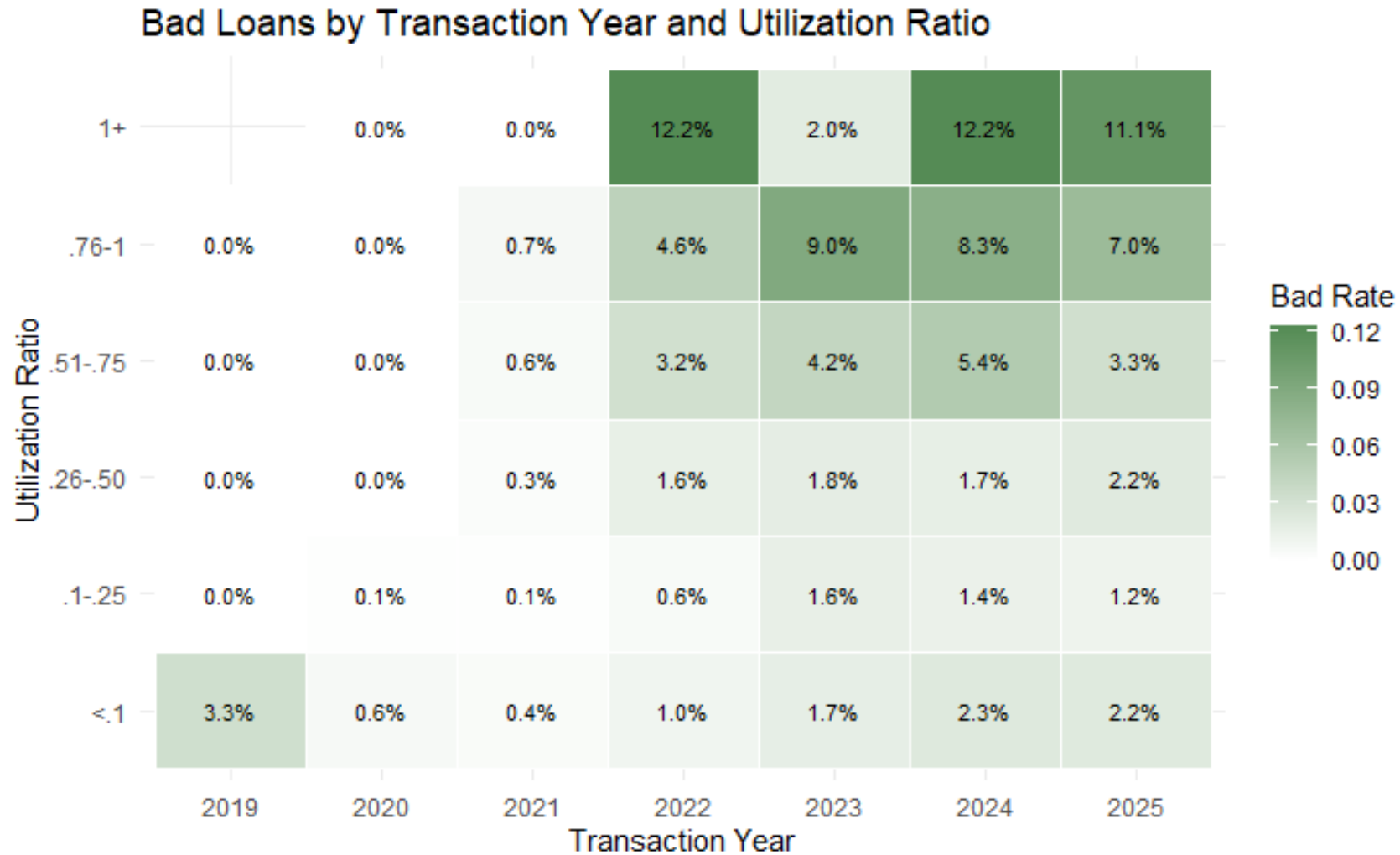
TRANSACTION YEAR



TRANSACTION YEAR AND HIGH UTILIZATION FLAG

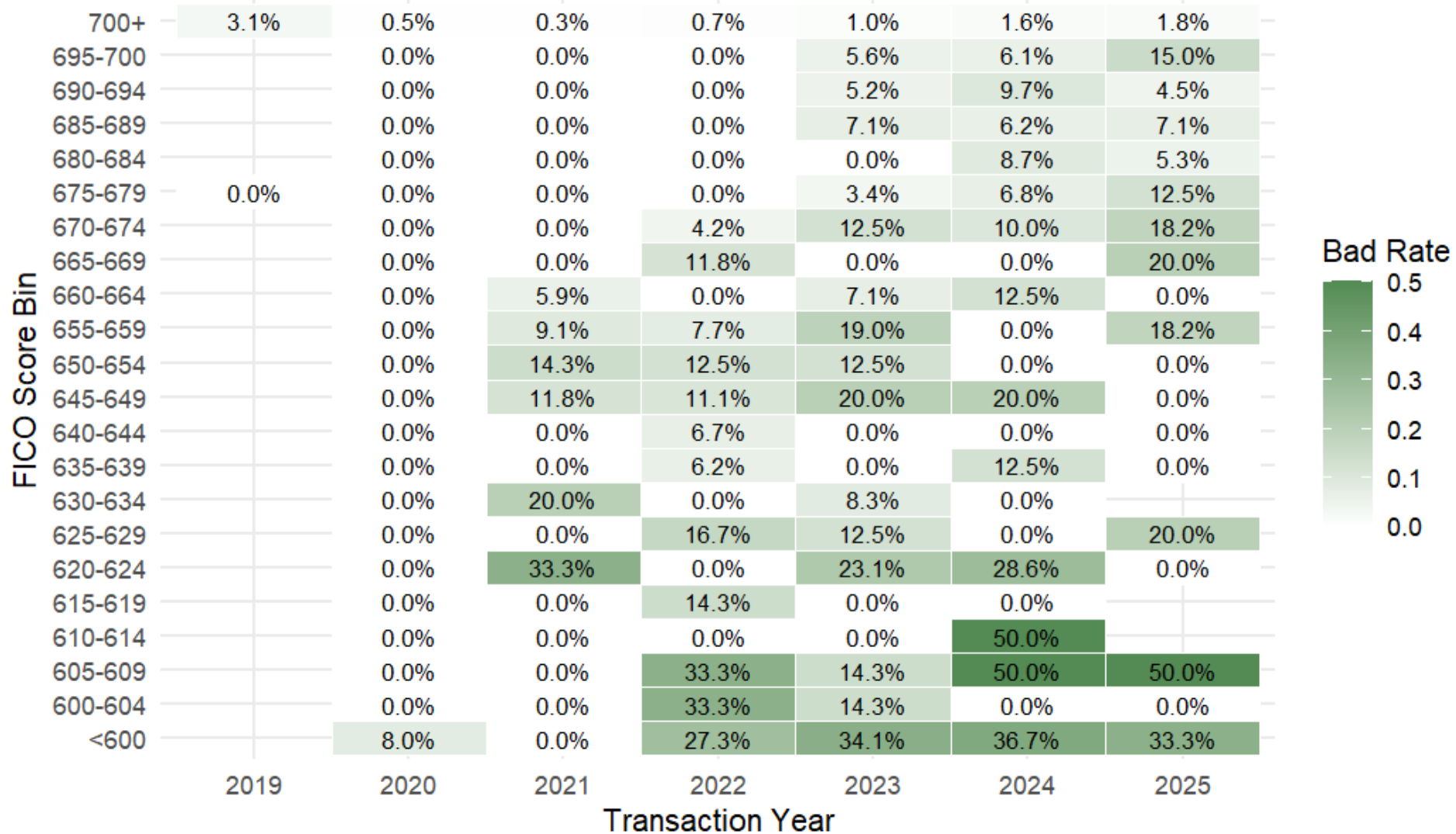


TRANSACTION YEAR AND UTILIZATION RATIO



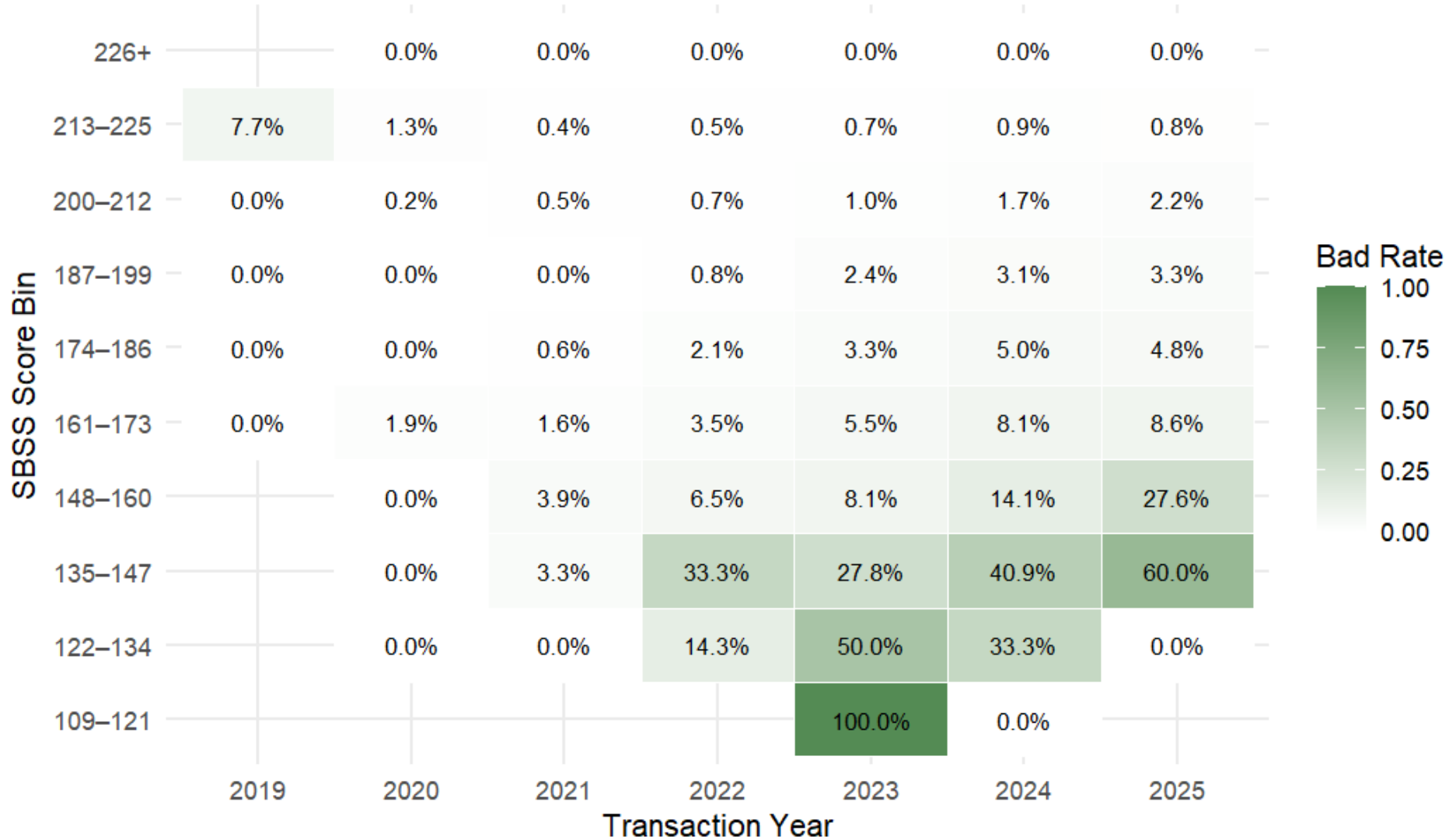
TRANSACTION YEAR AND FICO

Bad Loans by FICO Score Bin and Transaction Year



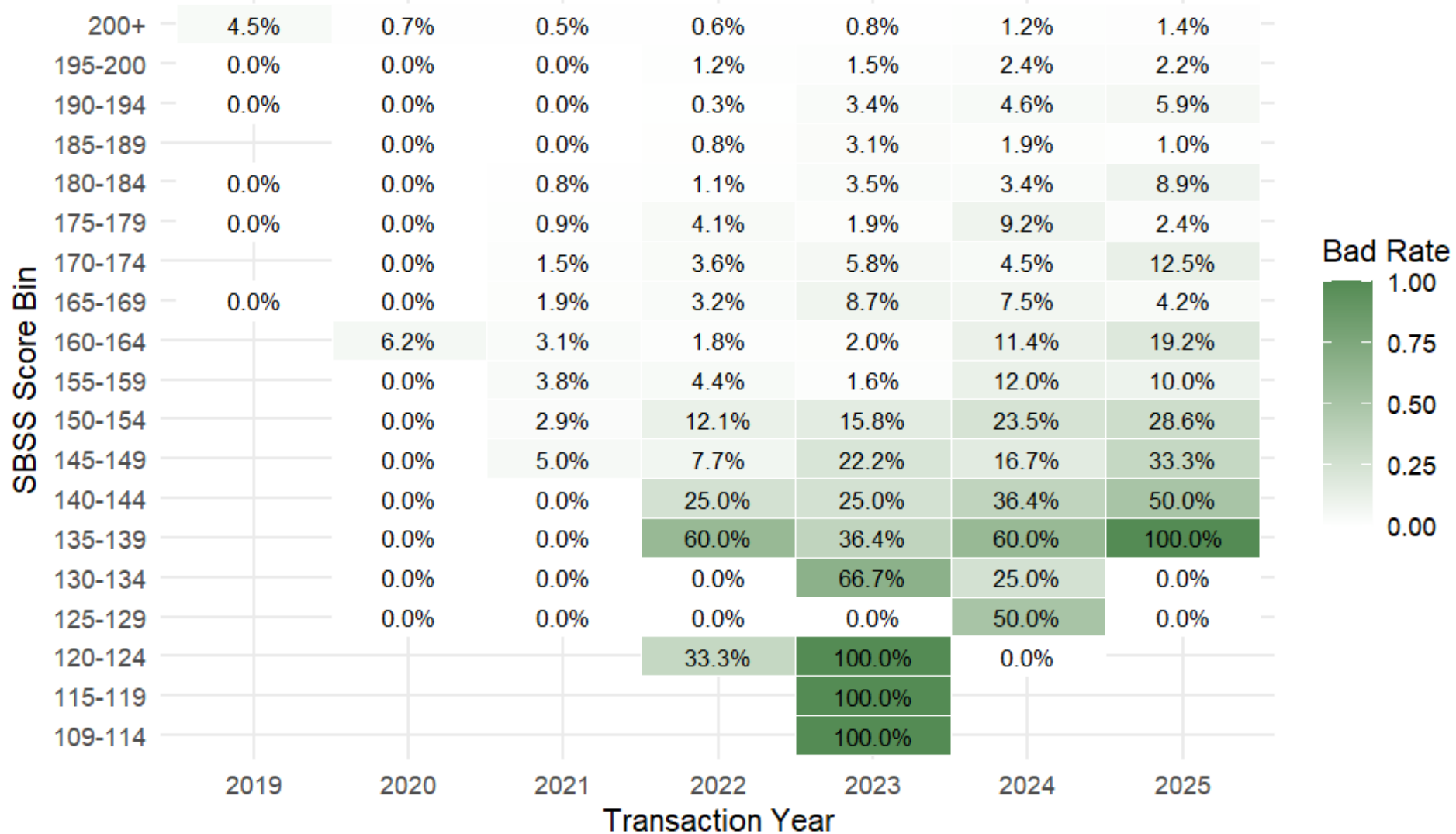
TRANSACTION YEAR AND SBSS

Bad Loans by SBSS Score Bin and Transaction Year



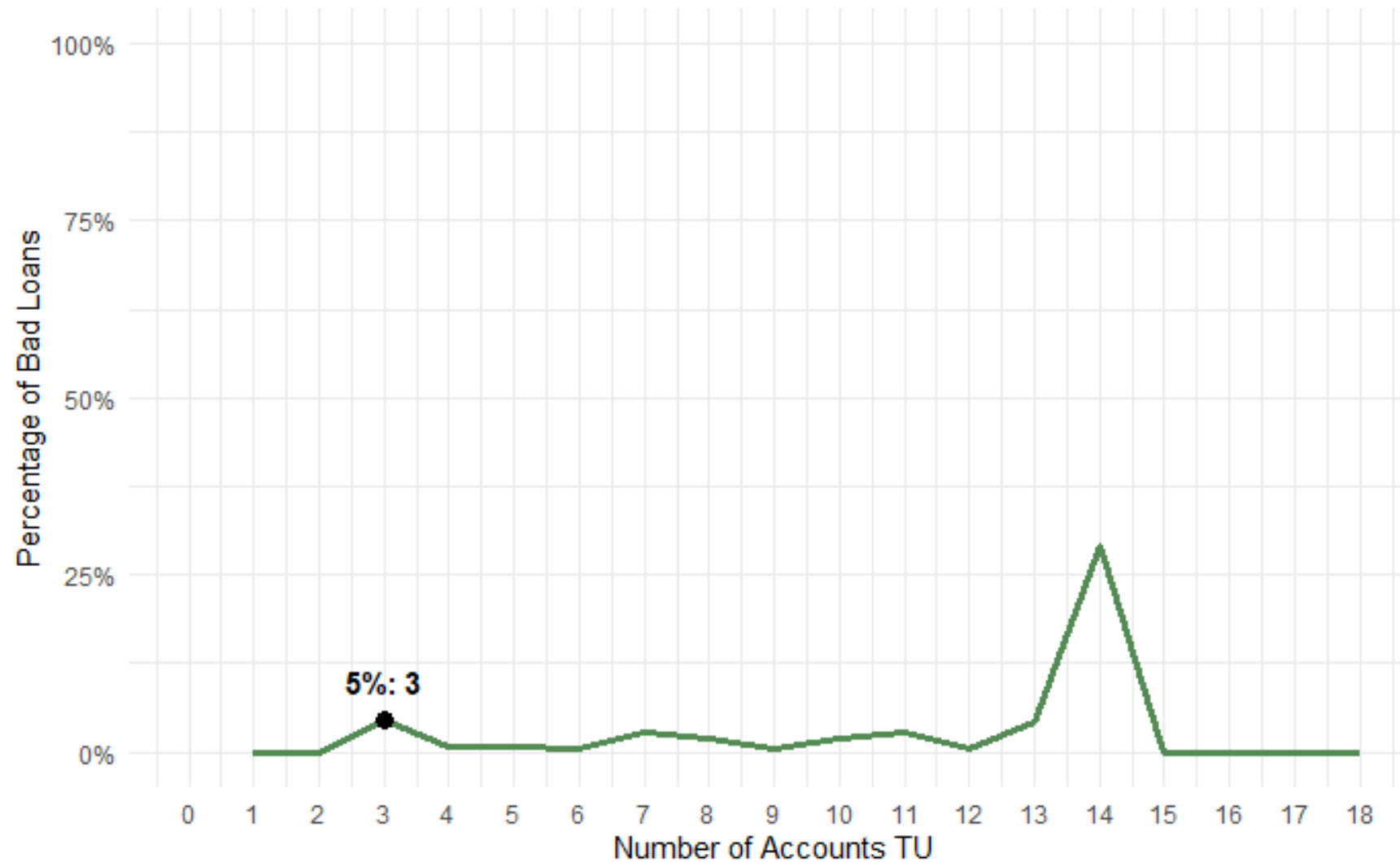
TRANSACTION YEAR AND SBSS

Bad Loans by SBSS Score Bin and Transaction Year





NUMBER OF TRANSUNION ACCOUNTS



NUMBER OF TRANSUNION ACCOUNTS

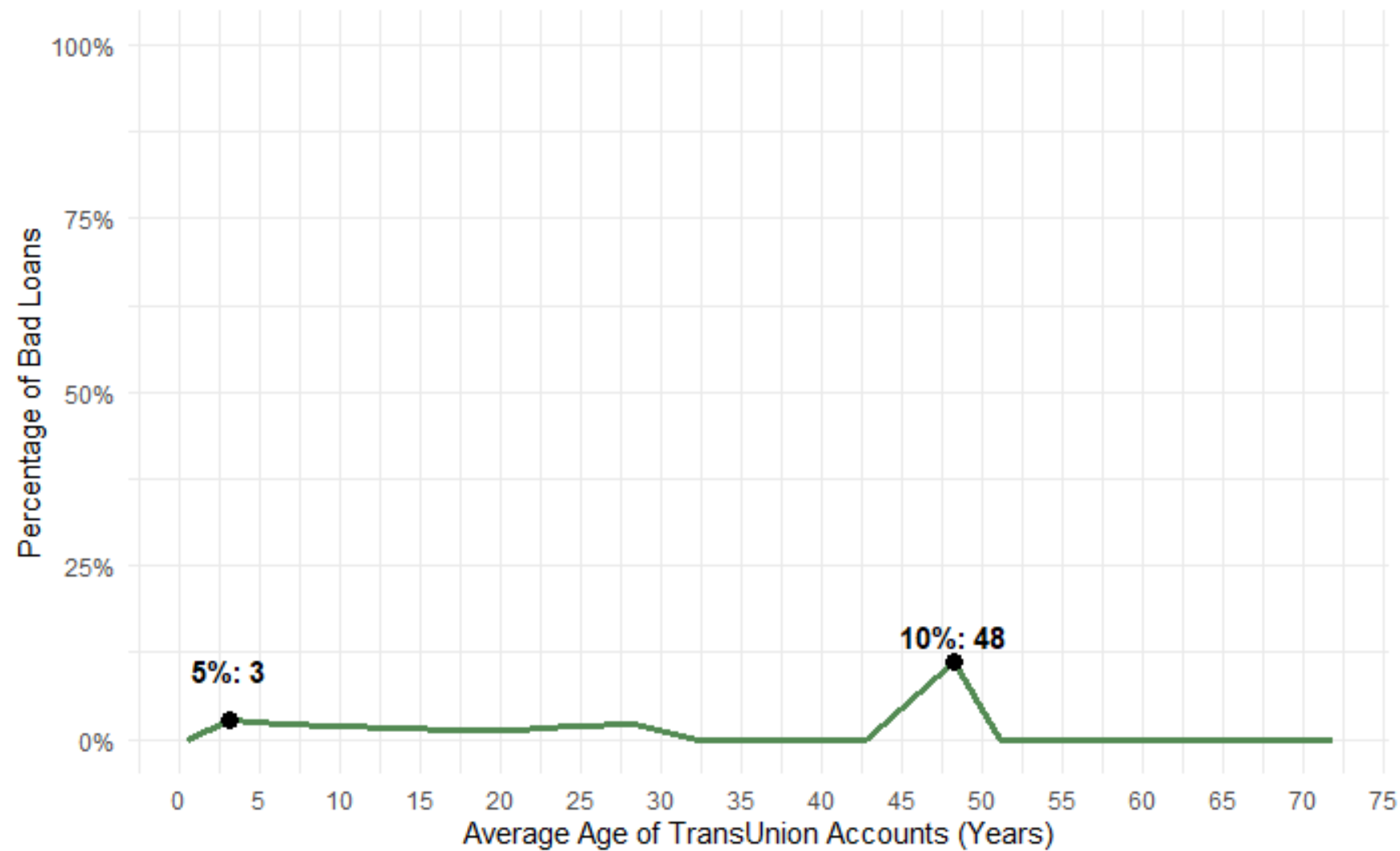
Number of TU Accounts	Count	Proportions of Obs	WOE	IV
1-2	1815	0.01	0.00	0.00
3-4	22545	0.07	0.12	0.57
5-6	65453	0.22	1.17	0.74
7-8	108023	0.36	-0.02	0.74
9-10	65521	0.22	0.34	0.77
11-12	29023	0.10	0.26	0.01
13-14	9800	0.03	-2.28	0.57
15-16	229	0.00	0.00	0.57
17-18	462	0.00	0.00	0.57

NUMBER OF TRANSUNION ACCOUNTS

Number of TU Accounts	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
1-2	1815	0	1815	0.00	302871	0.00
3-4	22545	355	22190	1.57	302871	0.12
5-6	65453	367	65086	0.56	302871	0.12
7-8	108023	1961	106062	1.82	302871	0.65
9-10	65521	833	64688	1.27	302871	0.28
11-12	29023	399	28624	1.37	302871	0.13
13-14	9800	1476	8324	15.06	302871	0.49
15-16	229	0	229	0.00	302871	0.00
17-18	462	0	462	0.00	302871	0.00



AVERAGE AGE OF TU ACCOUNTS



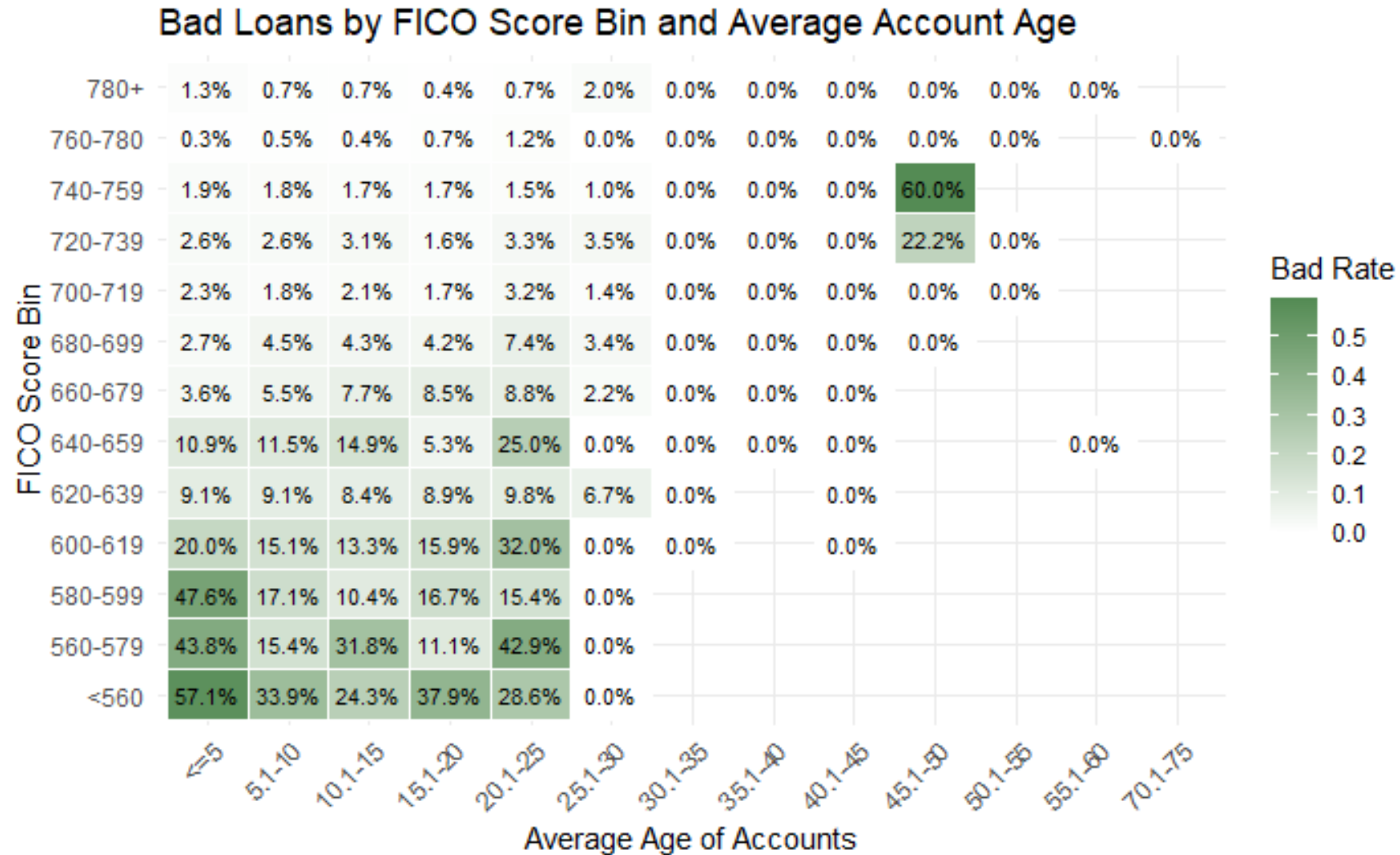
AVERAGE AGE OF TU ACCOUNTS TABLE (YEARS)

Average Age of TU Accounts	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
<=5	11981	409	11572	3.41	302871	0.14
5.1-10	103799	2189	101610	2.11	302871	0.72
10.1-15	109360	1820	107540	1.66	302871	0.60
15.1-20	45459	492	44967	1.08	302871	0.16
20.1-25	21448	378	21070	1.76	302871	0.12
25.1-30	7277	98	7179	1.35	302871	0.03
30.1-35	1808	0	1808	0.00	302871	0.00
35.1-40	1265	0	1265	0.00	302871	0.00
40.1-45	319	0	319	0.00	302871	0.00
45.1-50	72	5	67	6.94	302871	0.00
50.1-55	65	0	65	0.00	302871	0.00
55.1-60	17	0	17	0.00	302871	0.00
70.1-75	1	0	1	0.00	302871	0.00

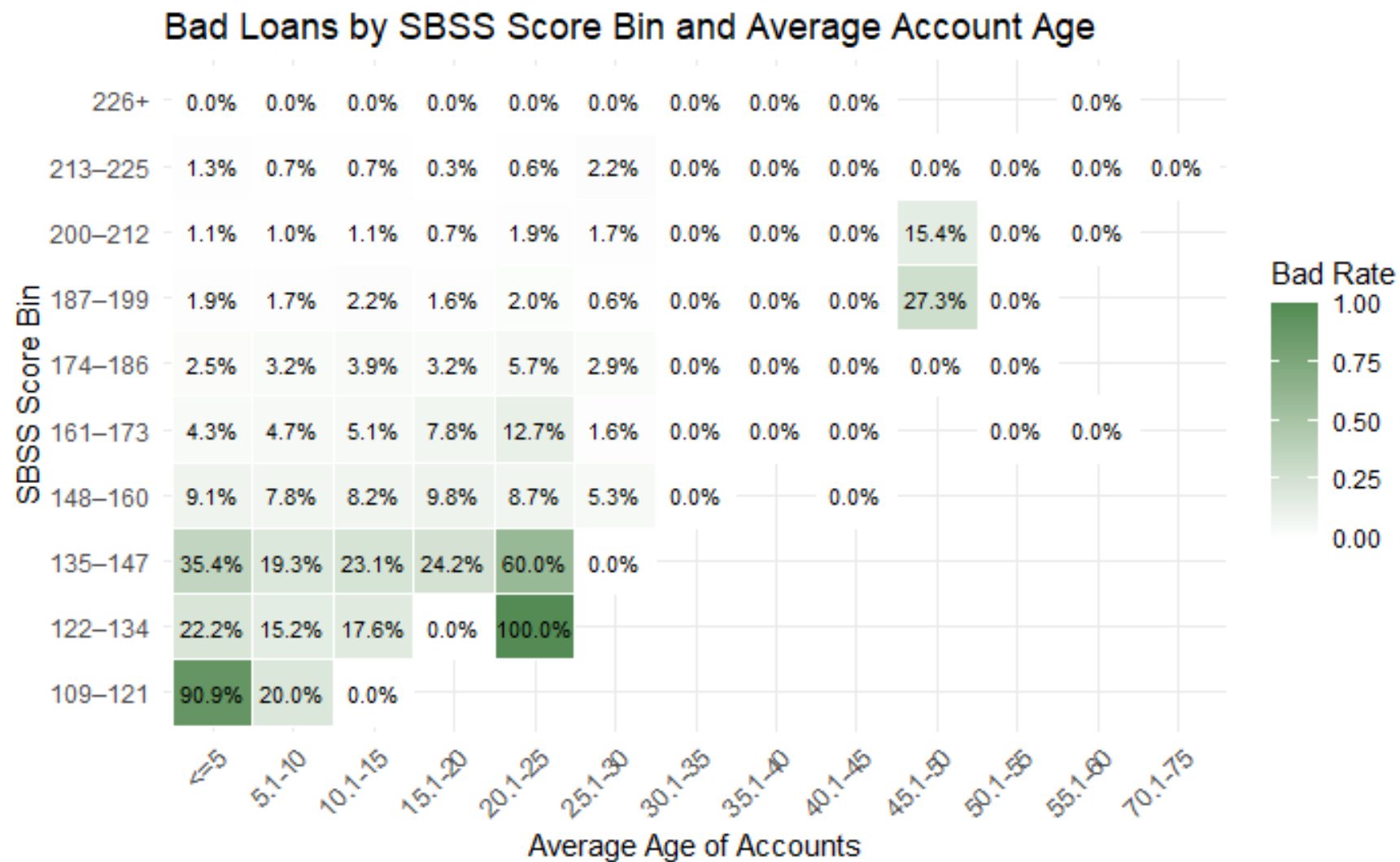
AVERAGE AGE OF TU ACCOUNTS TABLE (YEARS)

Average Age TU	Count	Proportions of Obs	WOE	IV
<=5	11981	0.04	-0.67	0.07
5.1-10	103799	0.34	-0.17	0.05
10.1-15	109360	0.36	0.07	0.00
15.1-20	45459	0.15	0.50	0.03
20.1-25	21448	0.07	0.01	0.03
25.1-30	7277	0.02	0.28	0.03
30.1-35	1808	0.01	0.00	0.03
35.1-40	1265	0.00	0.00	0.03
40.1-45	319	0.00	0.00	0.03
45.1-50	72	0.00	-1.42	0.03
50.1-55	65	0.00	0.00	0.05
55.1-60	17	0.00	0.00	0.05
70.1-75	1	0.00	0.00	0.05

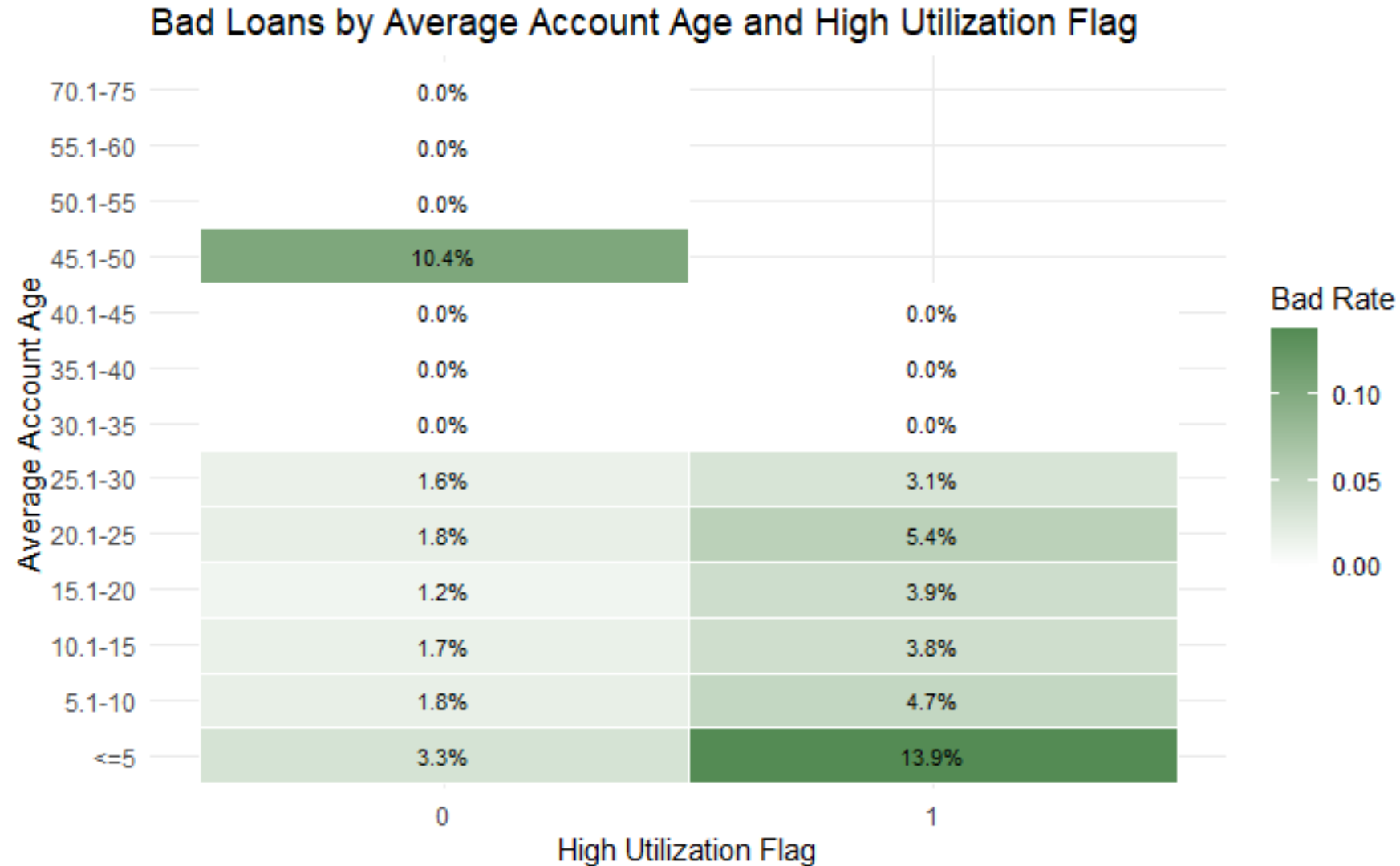
AVERAGE AGE OF TU ACCOUNTS AND FICO (YEARS)



AVERAGE AGE OF TU ACCOUNTS AND SBSS (YEARS)

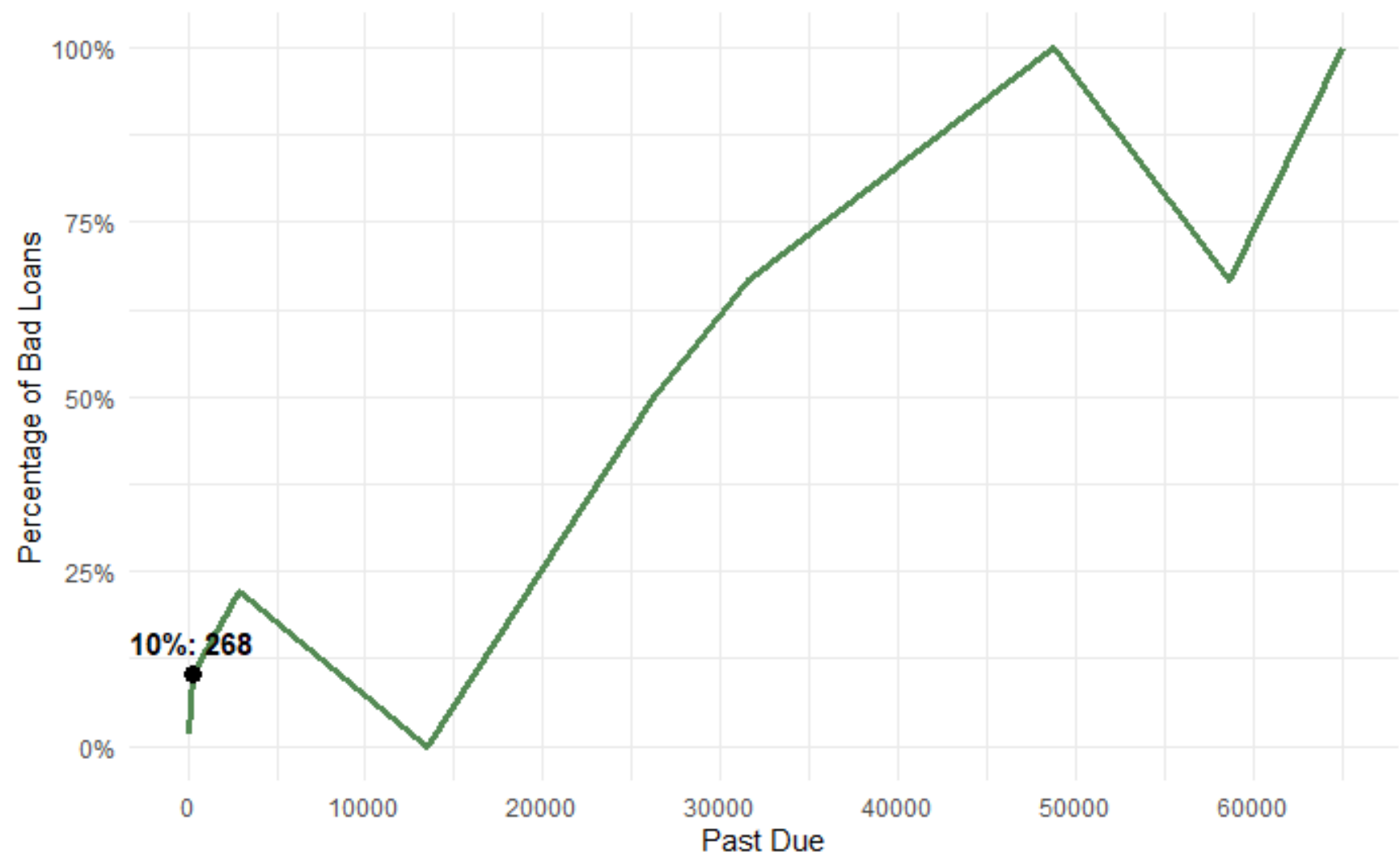


AVERAGE AGE OF TU ACCOUNTS AND UTILIZATION RATIO (YEARS)



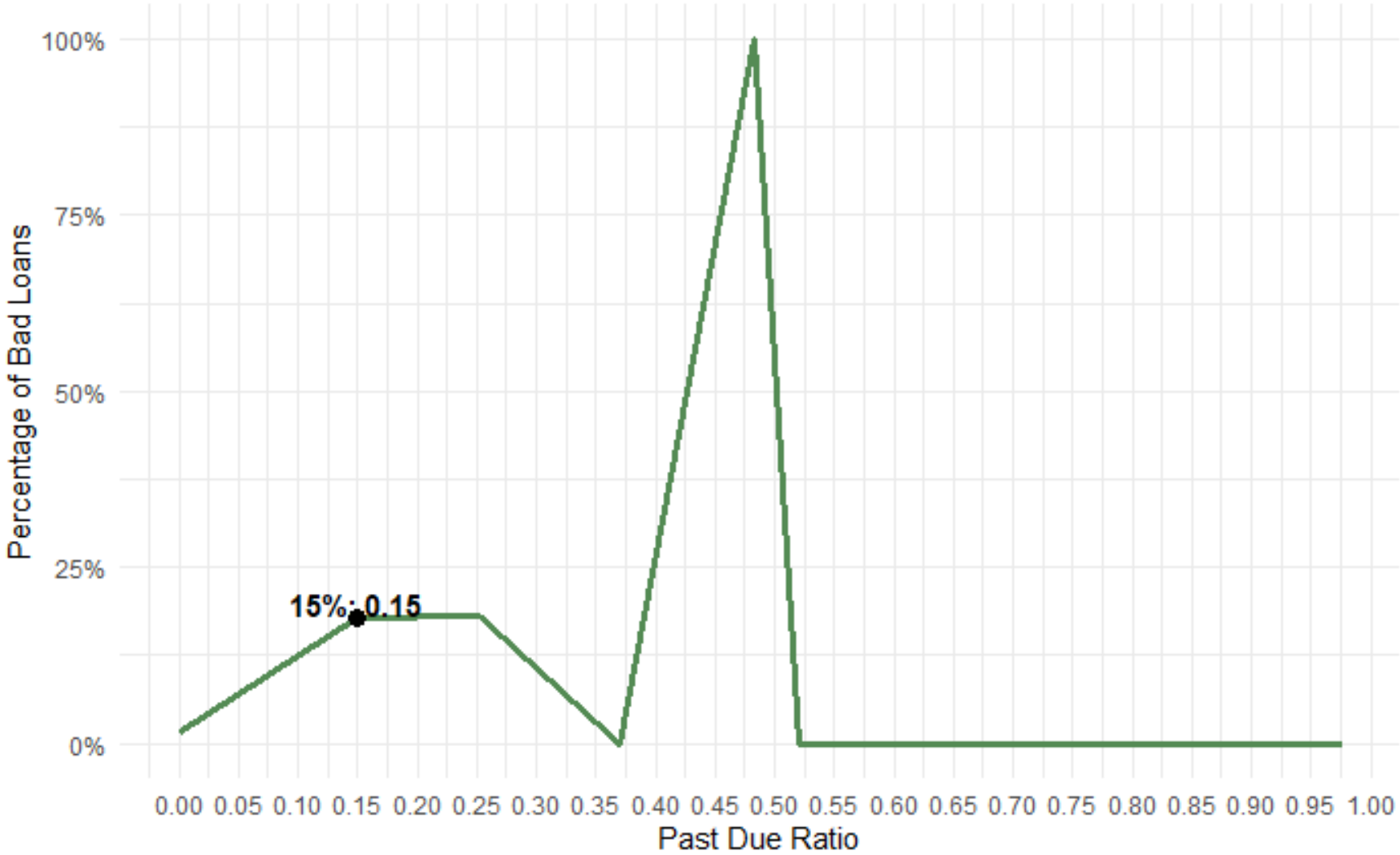


AMOUNT PAST DUE



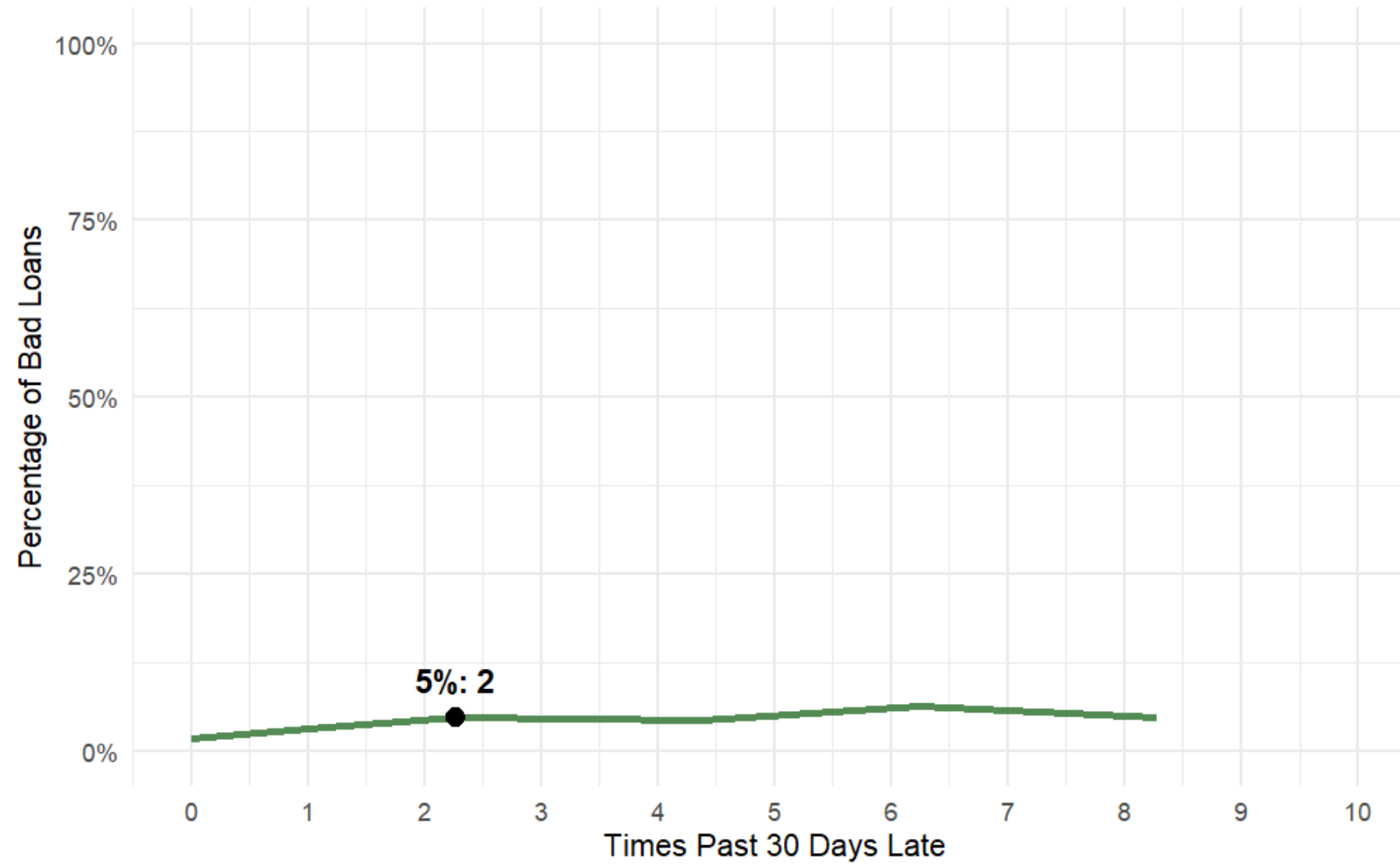


AMOUNT PAST DUE RATIO



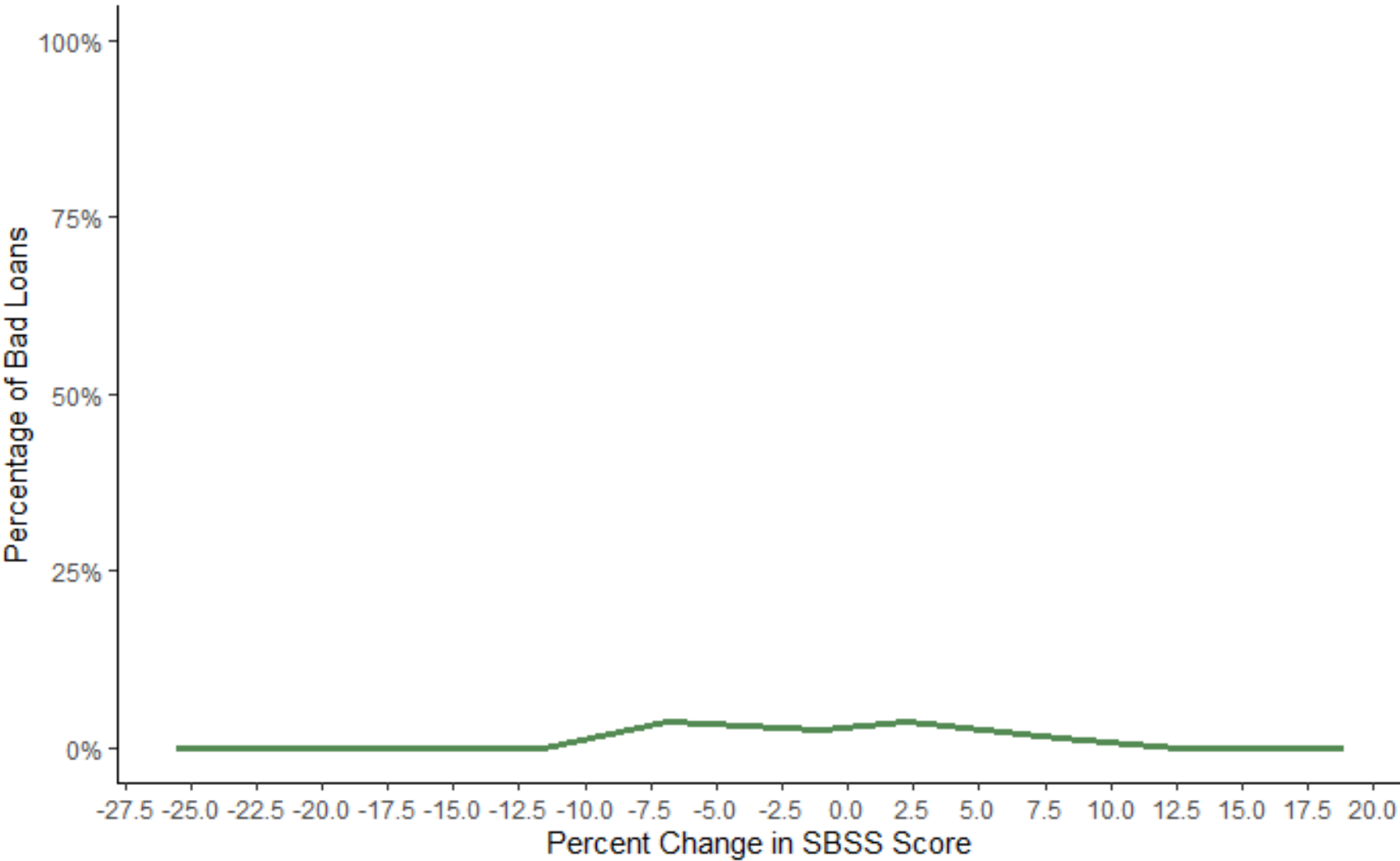


TIMES PAST 30 DAYS LATE



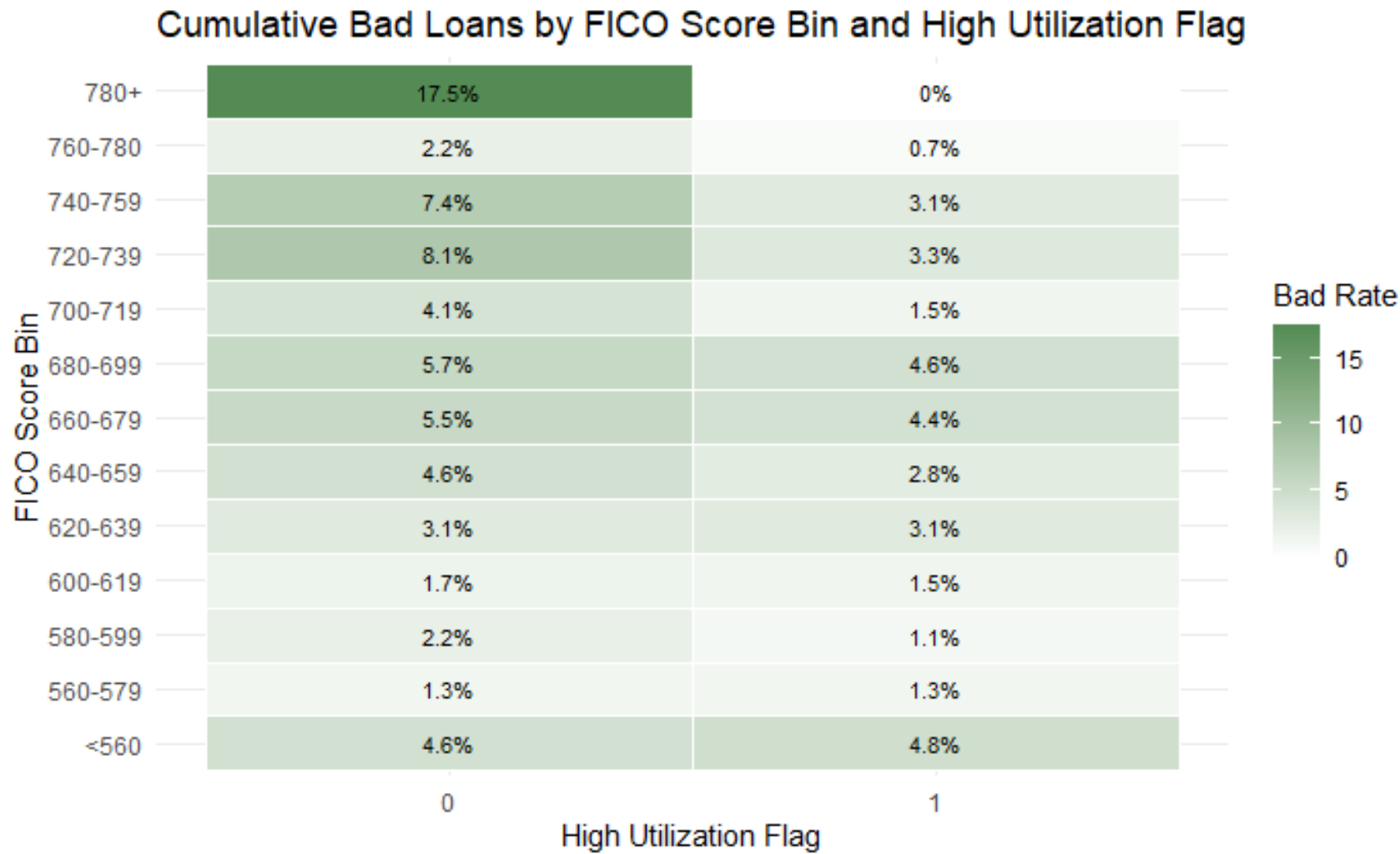


PERCENT CHANGE SBSS SCORE

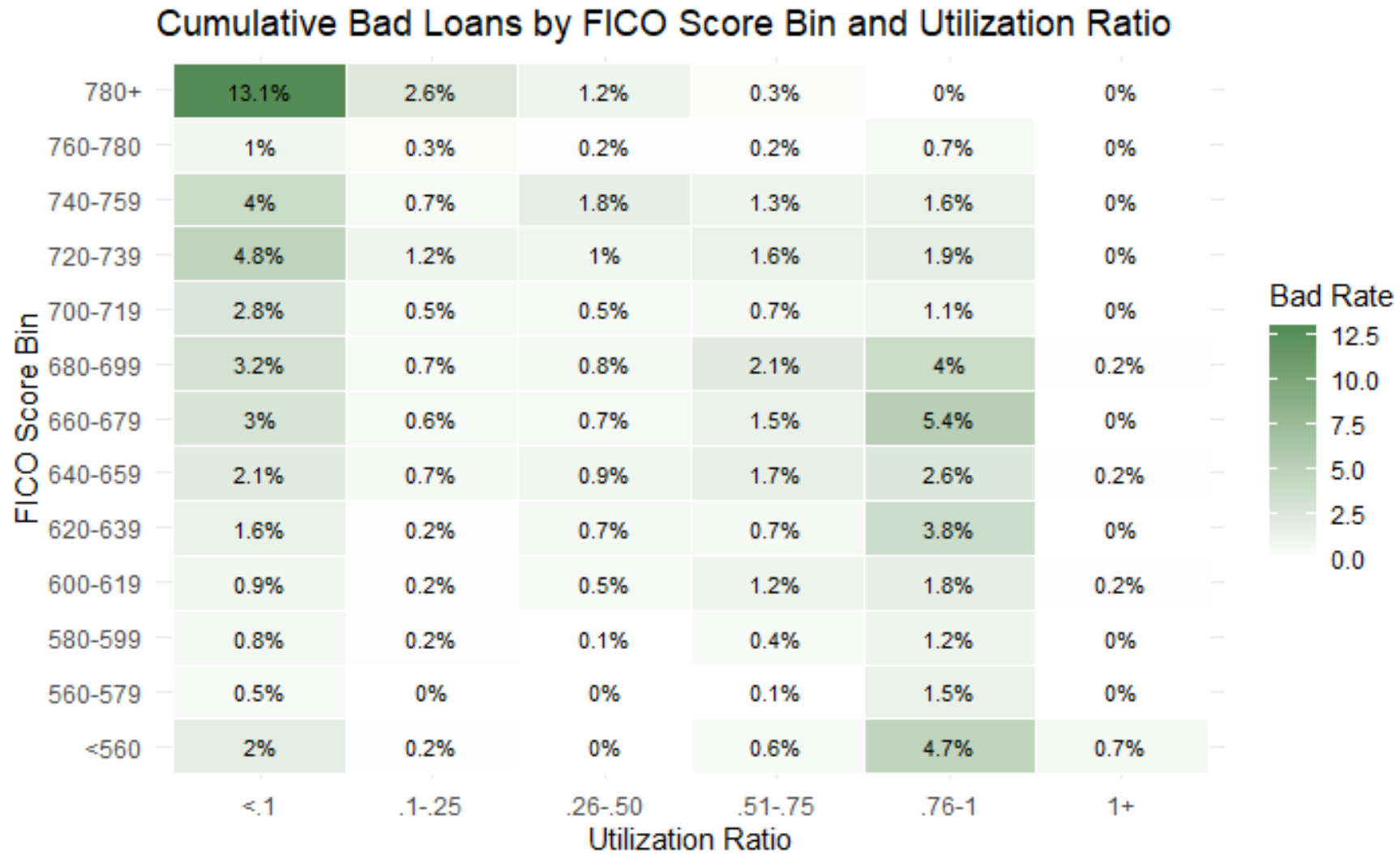


CUMULATIVE FICO AND HIGH UTILIZATION FLAG

- High Utilization Flag:**
- 0 = less than 80% credit usage
 - 1 = 80% credit usage or more



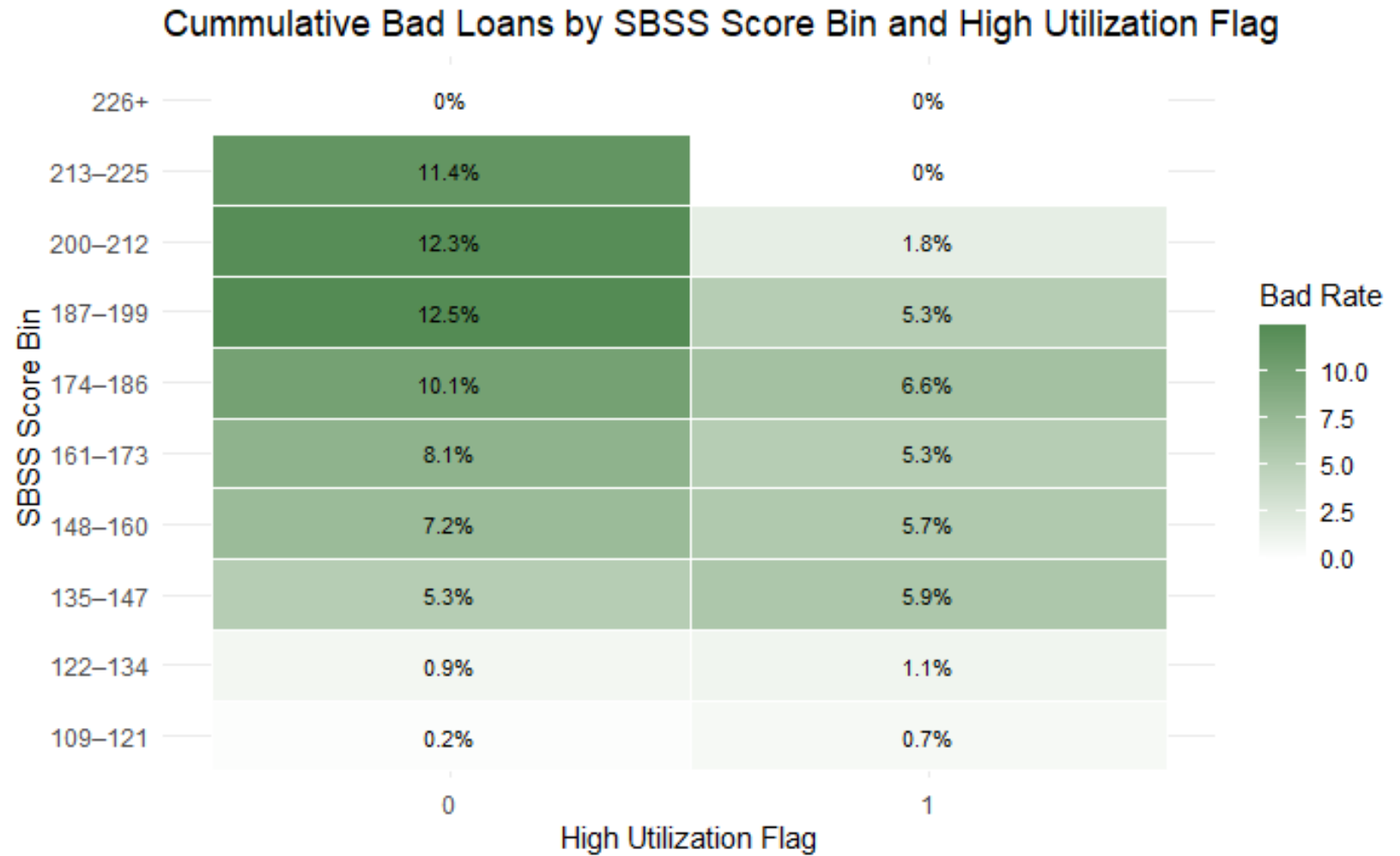
CUMULATIVE FICO AND UTILIZATION RATIO



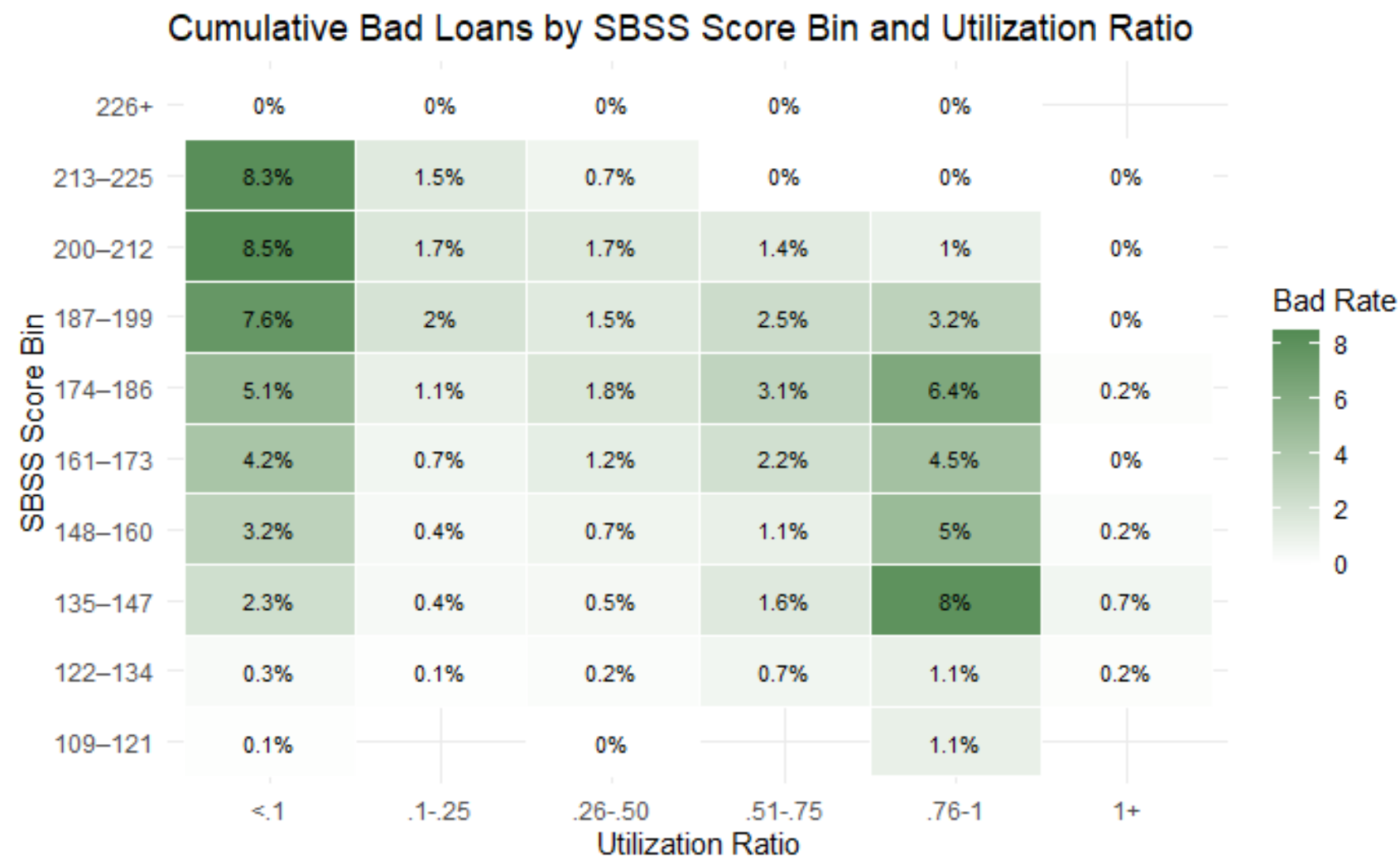
CUMULATIVE SBSS SCORE AND HIGH UTILIZATION

High Utilization Flag:

- 0 = less than 80% credit usage
- 1 = 80% credit usage or more



CUMULATIVE SBSS AND UTILIZATION RATIO

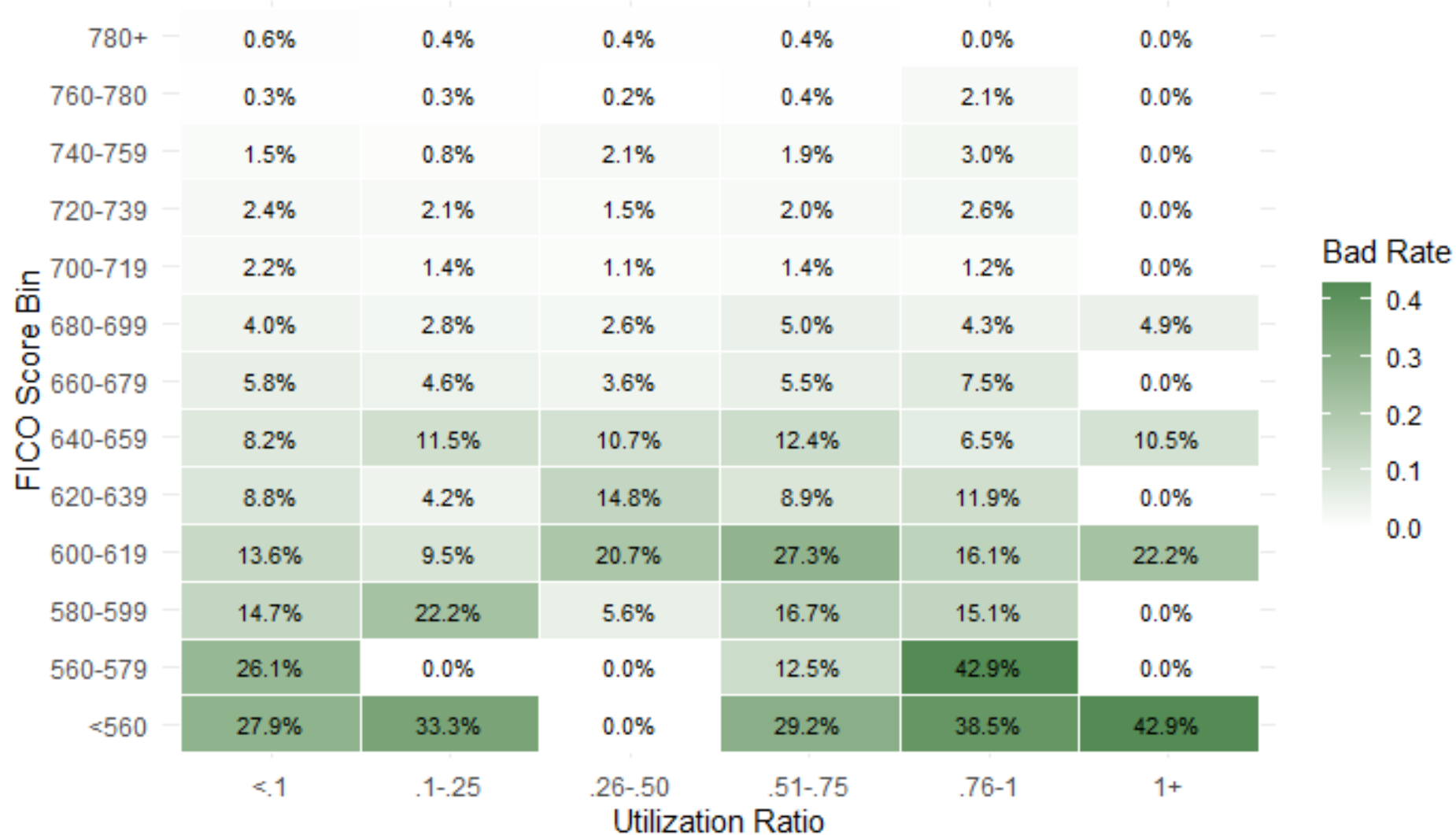


FICO TABLE

Fico Scores	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
<560	91	28	63	30.77	20760	0.13
560-579	25	7	18	28.00	20760	0.03
580-599	58	12	46	20.69	20760	0.06
600-619	73	9	64	12.33	20760	0.04
620-639	176	14	162	7.95	20760	0.07
640-659	266	23	243	8.65	20760	0.11
660-679	501	25	476	4.99	20760	0.12
680-699	782	26	756	3.32	20760	0.13
700-719	1206	19	1187	1.58	20760	0.09
720-739	1747	37	1710	2.12	20760	0.18
740-759	2074	34	2040	1.64	20760	0.16
760-780	2332	10	2322	0.43	20760	0.05
780+	11429	80	11349	0.70	20760	0.39

FICO AND UTILIZATION RATIO

Bad Loans by FICO Score Bin and Utilization Ratio

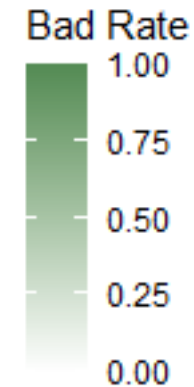


$$\text{Utilization Ratio} = \frac{\text{Credit Balance}}{\text{Credit Limit}}$$

FICO AND UTILIZATION RATIO

Bad Loans by FICO Score Bin and Utilization Ratio

FICO Score Bin	Utilization Ratio										
	<.1	.1-.2	.21-.3	.31-.4	.41-.5	.51-.6	.61-.7	.71-.8	.81-.9	.91-1	1+
700+	0.9%	0.6%	0.6%	0.6%	1.2%	0.8%	1.6%	1.4%	1.3%	2.3%	0.0%
695-700	4.4%	1.9%	0.0%	0.0%	3.0%	9.0%	4.8%	4.7%	2.6%	4.6%	9.1%
690-694	4.9%	4.0%	8.5%	2.4%	4.5%	4.9%	3.8%	7.5%	3.0%	5.1%	14.3%
685-689	4.0%	5.6%	0.0%	2.4%	2.0%	0.0%	5.6%	5.2%	8.4%	2.5%	0.0%
680-684	1.8%	0.0%	0.0%	3.7%	2.6%	3.4%	0.0%	1.9%	4.5%	4.5%	0.0%
675-679	3.5%	2.9%	6.2%	0.0%	6.7%	2.3%	12.8%	1.6%	2.7%	5.9%	0.0%
670-674	9.0%	8.1%	3.6%	10.3%	0.0%	4.8%	11.3%	7.1%	11.6%	19.2%	0.0%
665-669	6.7%	0.0%	0.0%	5.6%	0.0%	0.0%	0.0%	6.2%	2.4%	2.9%	0.0%
660-664	3.5%	3.6%	0.0%	0.0%	5.3%	0.0%	0.0%	3.1%	4.9%	3.1%	0.0%
655-659	9.0%	8.3%	13.3%	10.0%	0.0%	5.3%	8.0%	4.5%	2.9%	3.2%	0.0%
650-654	8.1%	13.3%	7.7%	25.0%	6.7%	21.4%	8.3%	0.0%	3.6%	5.5%	0.0%
645-649	11.1%	26.7%	18.2%	27.3%	0.0%	20.0%	22.7%	22.2%	17.4%	14.8%	0.0%
640-644	1.8%	0.0%	0.0%	0.0%	11.1%	0.0%	0.0%	16.7%	8.7%	2.5%	66.7%
635-639	2.8%	0.0%	5.6%	11.1%	0.0%	0.0%	14.3%	0.0%	2.6%	5.2%	0.0%
630-634	6.4%	0.0%	0.0%	50.0%	20.0%	16.7%	20.0%	5.6%	6.7%	8.2%	0.0%
625-629	7.4%	0.0%	0.0%	11.1%	25.0%	0.0%	8.3%	0.0%	18.2%	17.9%	0.0%
620-624	23.8%	20.0%	33.3%	20.0%	20.0%	11.1%	16.7%	38.5%	26.3%	16.7%	0.0%
615-619	9.5%	14.3%	100.0%	33.3%	16.7%	25.0%	60.0%	40.0%	11.1%	4.0%	0.0%
610-614	5.6%	33.3%	0.0%		0.0%	0.0%	50.0%	20.0%	0.0%	4.3%	0.0%
605-609	19.0%	0.0%	0.0%	0.0%	20.0%	12.5%	28.6%	20.0%	45.5%	27.3%	100.0%
600-604	19.0%	0.0%	0.0%	0.0%	33.3%	0.0%	57.1%	50.0%	0.0%	16.0%	0.0%
<600	22.6%	25.0%	20.0%	0.0%	0.0%	33.3%	20.6%	20.0%	31.6%	31.9%	36.0%

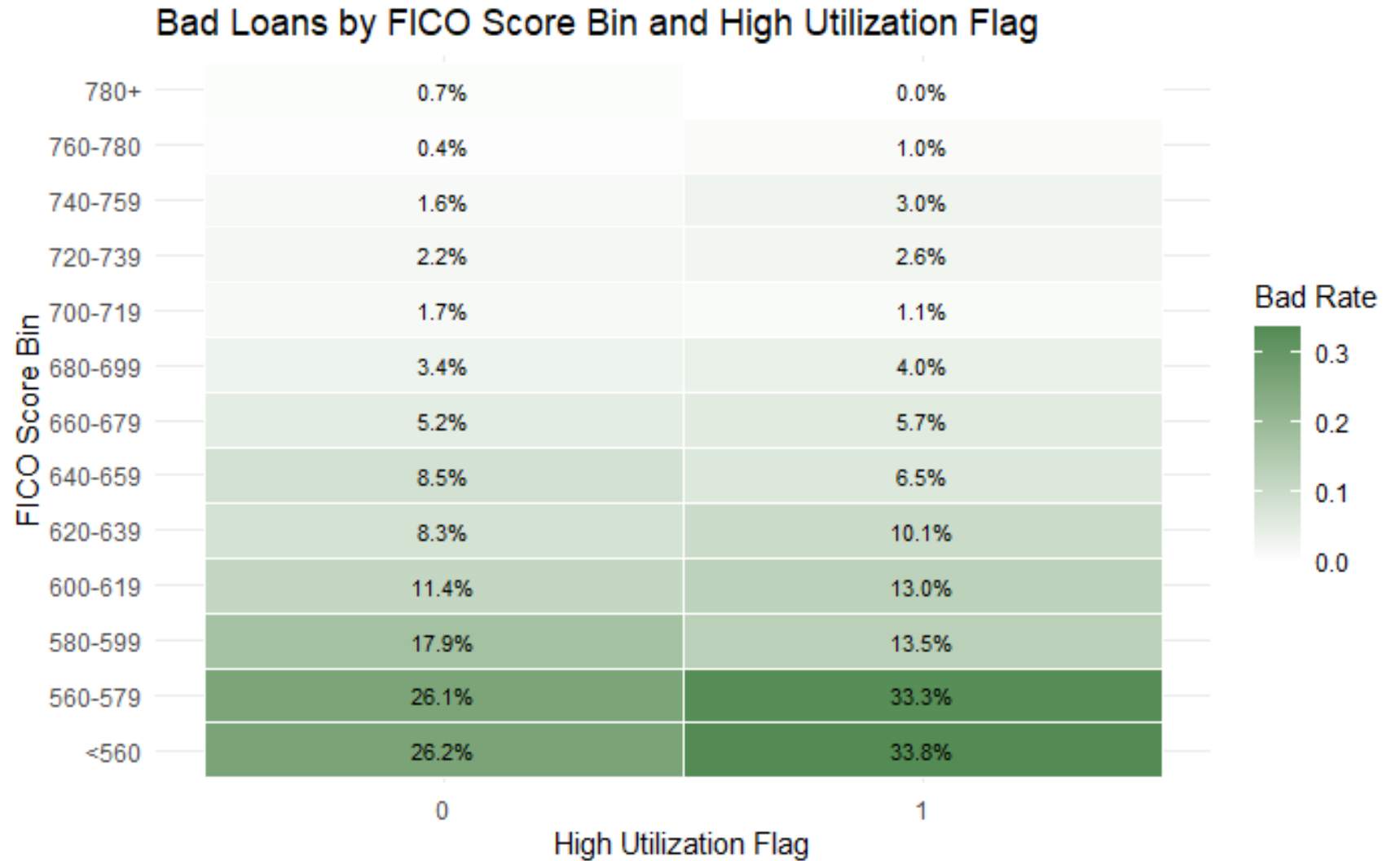


$$\text{Utilization Ratio} = \frac{\text{Credit Balance}}{\text{Credit Limit}}$$

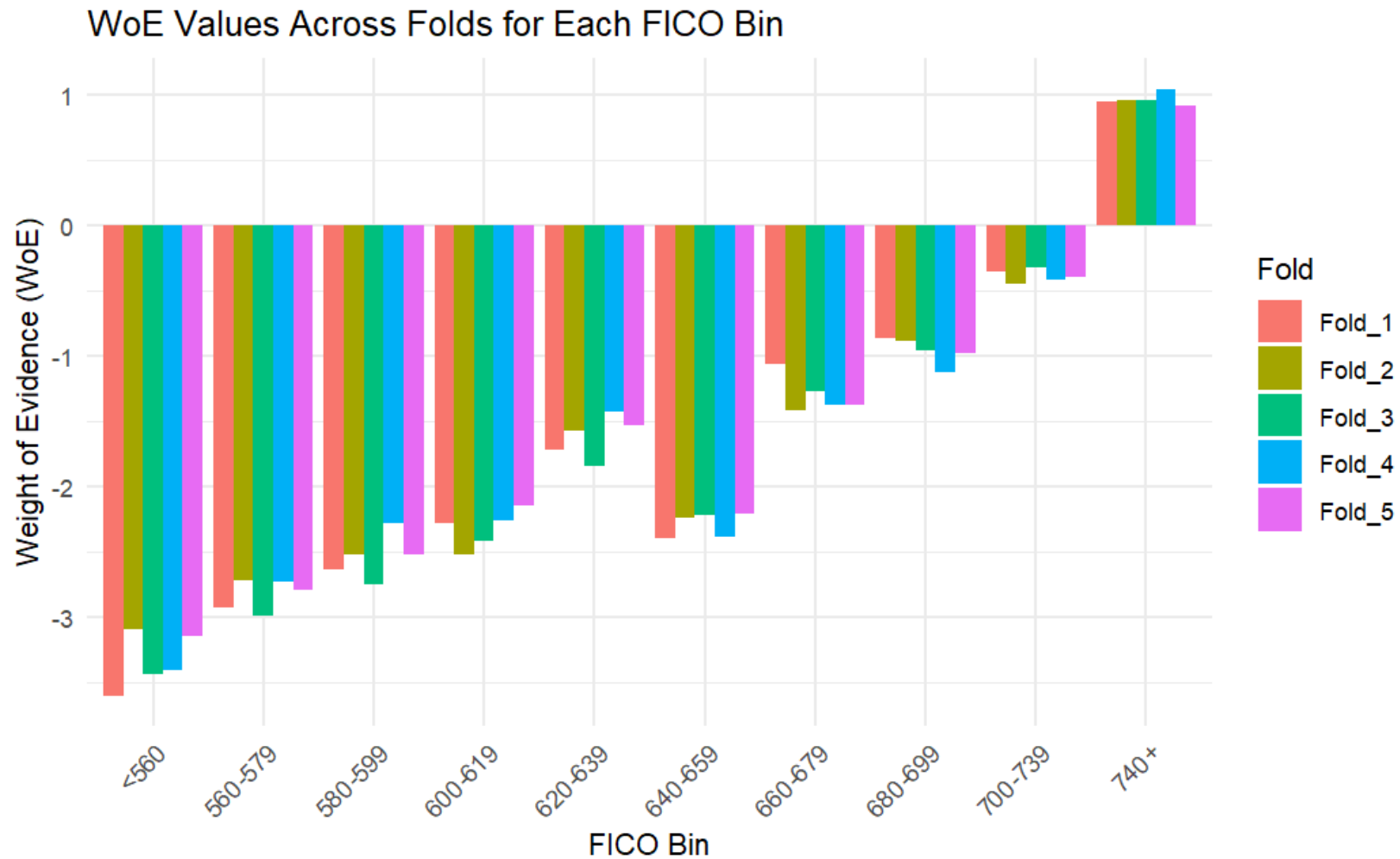
FICO AND HIGH UTILIZATION FLAG

High Utilization Flag:

- 0 = less than 80% credit usage
- 1 = 80% credit usage or more



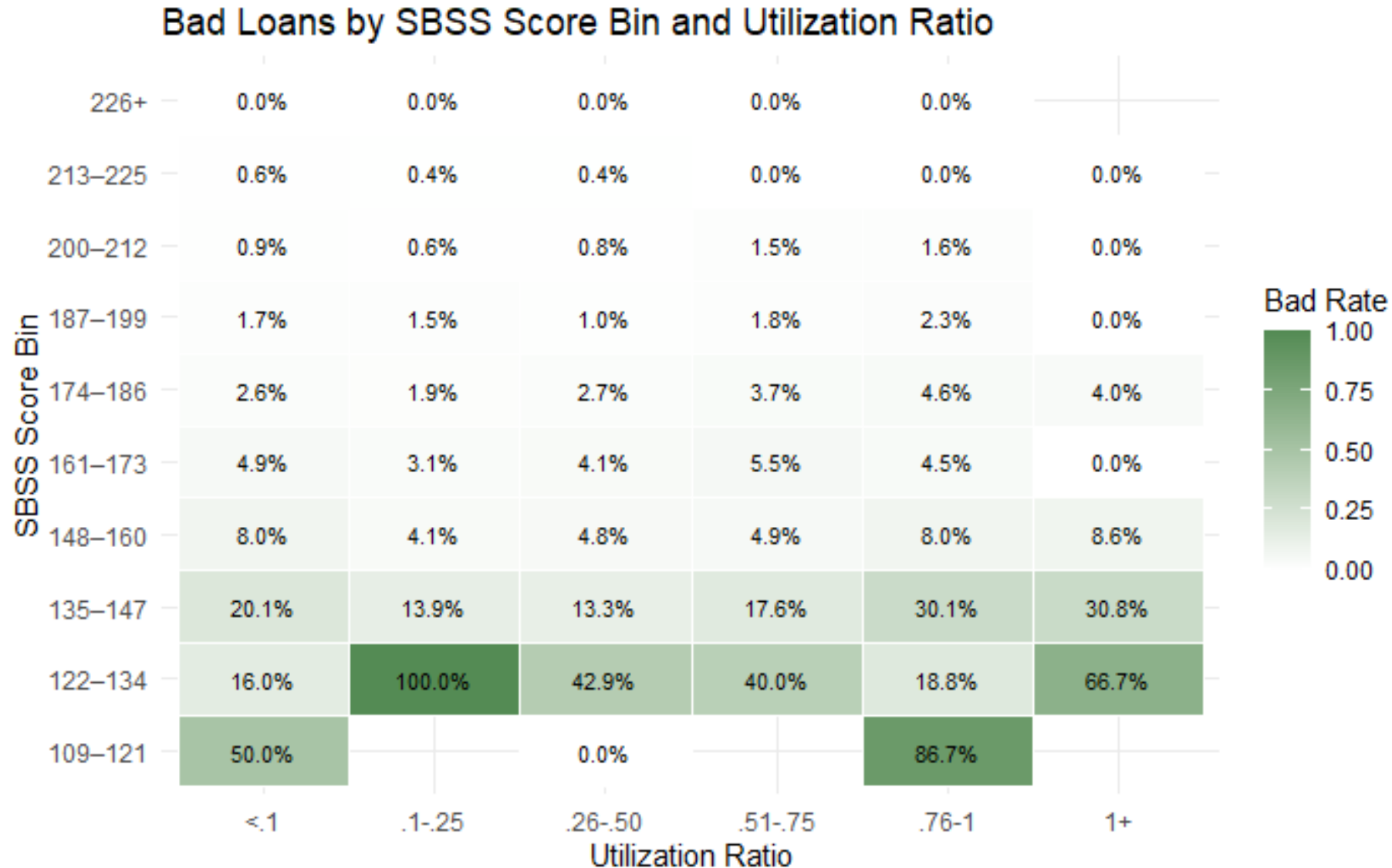
STABILITY OF WOE FOR FICO BINS



SBSS TABLE

SBSS Scores	Loans (Count)	Bads (Count)	Goods (Count)	Bads %	Total Loans	Bads % (All Loans)
109–121	4	3	1	75.00	20382	0.01
122–134	34	5	29	14.71	20382	0.02
135–147	135	29	106	21.48	20382	0.14
148–160	435	36	399	8.28	20382	0.18
161–173	836	38	798	4.55	20382	0.19
174–186	1880	47	1833	2.50	20382	0.23
187–199	3697	57	3640	1.54	20382	0.28
200–212	5868	56	5812	0.95	20382	0.27
213–225	7408	52	7356	0.70	20382	0.26
226+	85	0	85	0.00	20382	0.00

SBSS AND UTILIZATION RATIO

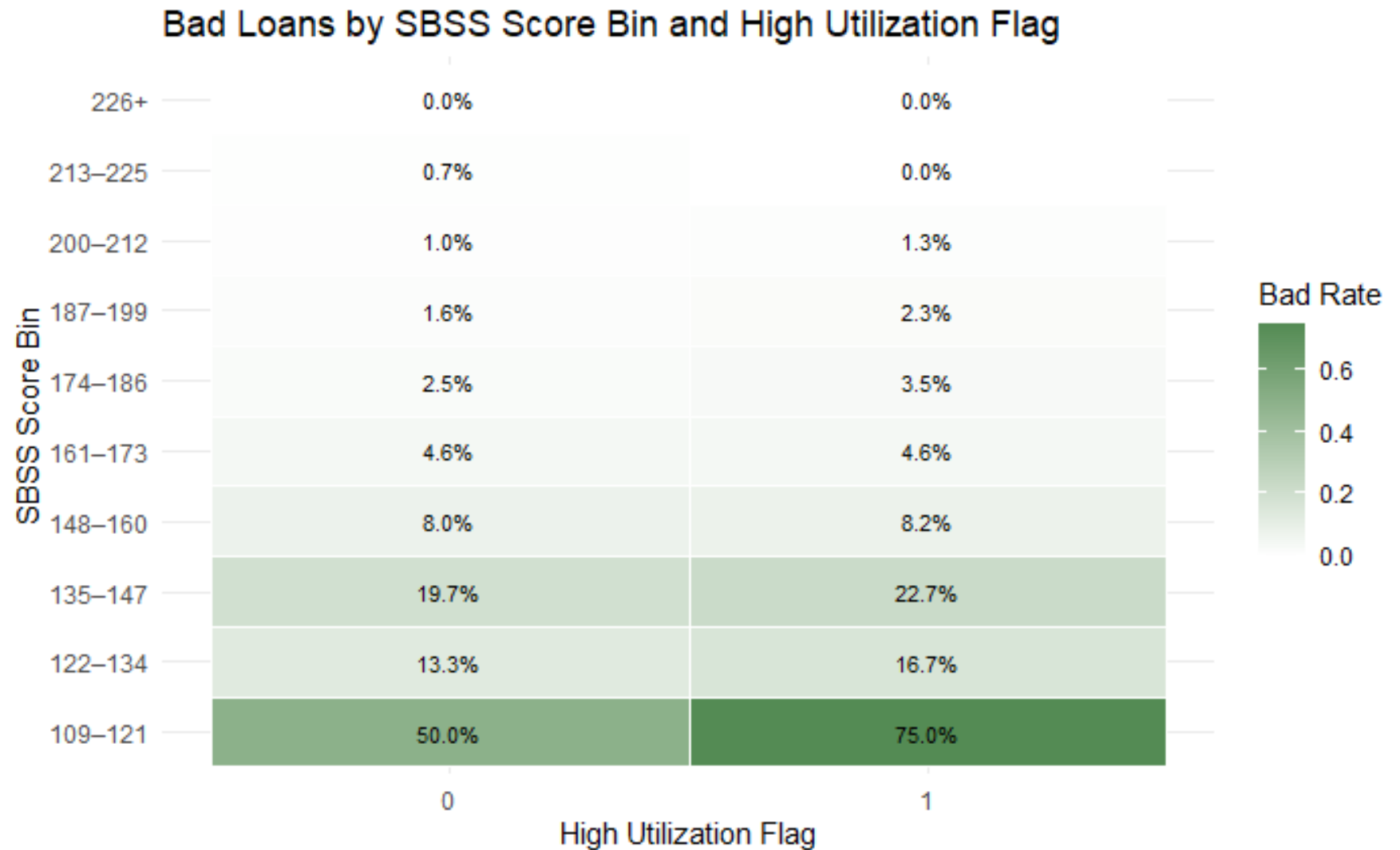


$$\text{Utilization Ratio} = \frac{\text{Credit Balance}}{\text{Credit Limit}}$$

SBSS AND HIGH UTILIZATION FLAG

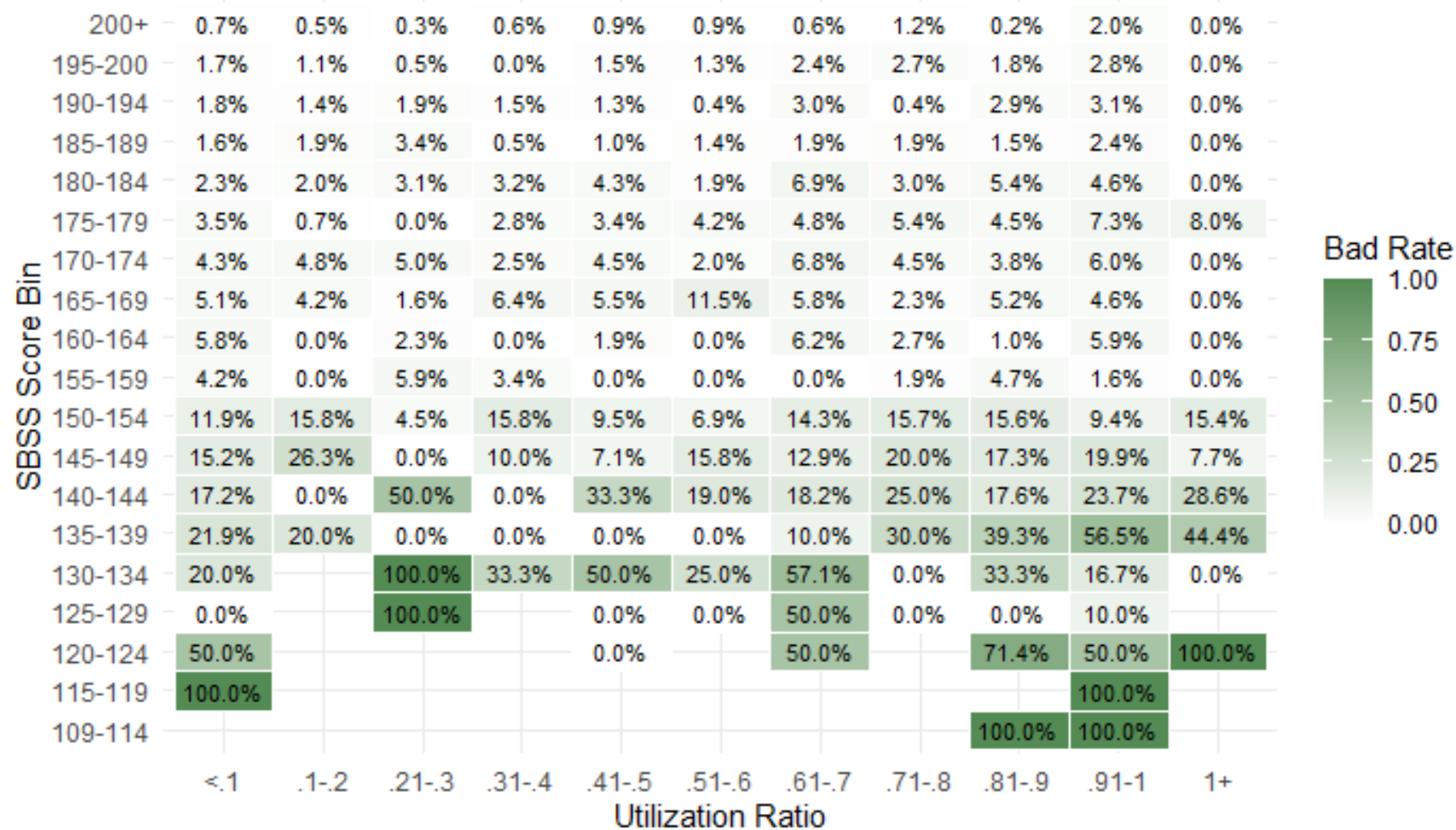
High Utilization Flag:

- 0 = less than 80% credit usage
- 1 = 80% credit usage or more



SBSS AND UTILIZATION RATIO

Bad Loans by SBSS Score Bin and Utilization Ratio



$$\text{Utilization Ratio} = \frac{\text{Credit Balance}}{\text{Credit Limit}}$$